



P A C K E T A D E

Midway Review

Six-Month Checkpoint – Flagship Deliverable

Full multi-fleet review of the PacketADE Agent Development Environment

7 domain deep dives · 22-agent bug hunt (16 confirmed, all fixed)

Competitive & technology briefs · 4-lens vision panel · 90-day plan

Expanded same day: UX ledger (43 findings, 5 owner decisions) · visual audit · 64-doc audit sweep

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1. Executive Summary

At its six-month midpoint, PacketADE is a real product, not a prototype. The engineering quality in its hardest subsystems — atomic state persistence with corruption recovery, a concurrency-correct agent turn lifecycle, battle-hardened xterm/PTY handling, a versioned sidecar protocol with restart supervision, and above-average SSH security — is well beyond what solo desktop projects usually achieve. Its breadth is its moat: an eight-provider chat matrix mixing subscription OAuth with API keys and local models, SSH-remote execution for terminals and agent tools, worktree-isolated parallel Flights with cost rollup, a memory layer, an MCP hub, and Windows-native support. No competitor bundles all of this.

Weaknesses cluster in three bands, none of them structural rot:

- **Lifecycle and recovery seams.** Both criticals and most highs lived where sessions cancel, error, crash, or restart — the seams the test suites never exercised against real providers.
- **Process infrastructure.** Zero CI, no coverage measurement, unsigned installers, no auto-updater, no crash telemetry, and macOS/Linux targets never compiled. The quality ladder is good — and entirely voluntary.
- **Daily-driver polish.** Ctrl+K stealing keystrokes from terminals, forgotten view on every launch, one-click session-destroying close, ~96 silent failure sites, a flagship launch modal that forgets its targets.

~177k	16 / 0	2 · 9 · 5	16 of 16	8	0
LOC under review	bugs confirmed / refuted	critical · high · medium	bugs fixed this session	chat providers, one contract	CI workflows / signed builds

Headline strengths

- Transport-agnostic api-agent:* event contract spanning the in-process runtime and the Node sidecar.
- Durable-by-design persistence: atomic writes, quarantine-first recovery, backend-owned merge semantics.
- SSH posture: host-key pinning, sh_quote everywhere, keyring passwords via askpass, fail-closed containment.
- Test depth where risk is highest: 165 Vitest files, 417 Rust tests, 14 sidecar smoke gates, a schema drift gate.
- Flights: one-prompt-N-agents with collision gating, Rust-enforced reviewer gates, per-flight cost rollup.

Headline risks (post-fix)

- Default "auto" mode maps to --dangerously-bypass-approvals-and-sandbox for Codex sessions (critical; open).
- No Flight recovery across restarts: zombie attempts, and failed attempts destroy their uncommitted work (open).
- Discipline-only quality ladder: at 177k LOC with no CI, one skipped gate ships regressions silently.
- Distribution stuck: unsigned, updater-less releases cap the user base at approximately one.
- Gated-write approvals hang forever on the flagship Claude subscription provider (pending_edit missing toolUsed — one-line fix, outside the 16-bug scope; open).

What was fixed this session

All 16 confirmed bugs repaired (15 distinct defects; two findings reconciled as one), with regression tests. Highlights: the Anthropic cancel-kills-conversation critical, Codex flight-cost over-counting critical, merge failures masquerading as success, the zombie-sidecar hang, session-bricking on advisory errors, and the localStorage migration ordering defect. Also: Gemini CLI removed (45 files), docs rewritten for honesty, statusline shipped in a sibling repo, marketing identity produced.

Same-day expansion — the consolidated six-month ledger

This report was expanded the same day with three reconciliation inputs. (1) **The UX Ledger (Chapter 3)**: the 07-29 main-shell audit, this review's code critique, and a rendered visual audit reconciled into **43 deduped findings** (9 critical, 19 high, 13 medium, 2 low) — **none** resolved by this session's 16 bug fixes, which were all store/backend defects. At the front of the queue sit **five owner decisions**: minutes of owner time that unblock every MS1-MS4 implementation wave — the top pending calls in the whole review. (2) **Visual audit (Chapter 4)**: the first rendered-pixel pass, 14 curated screenshots with per-screen findings and an honest web-mode coverage boundary. (3) **Outstanding audits ledger (Chapter 10)**: a validity sweep of all **64** past audits/reviews/scans/plans: **218 open items**, **182 still valid** as of today, **15 critical**.

Verdict: the project has earned a "ship it to more humans" posture — after the trust seams close, starting with the five owner decisions, in the order of Section 11.

2. State of the Codebase

Seven domain deep dives. Each chapter: reviewer summary, selected strengths, ranked problems (severity-tagged), recommendations. Problems fixed in this session's bug loop are marked [FIXED]; everything else is open.

2.1 Frontend — React 19 / TypeScript / Zustand

A mature, unusually well-documented codebase (~119k LOC in src/) with sophisticated engineering in its hardest areas: the rAF-coalesced streaming pipeline with flush-before-done ordering; xterm/PTY handling working around an xterm 6.0 parser crash, WebGL blanking, and StrictMode double-spawns; and disciplined hydration with sessionGlue as the single eslint-enforced bridge between engines. 50+ store test suites; inline design rationale makes a 55-store architecture legible.

Weaknesses: state-management scale effects (one conversations array holds every transcript; always-mounted chrome recomputes O(all-messages) cost per streaming frame; unmemoized rows re-parse markdown per frame); persistence fragmented across five mechanisms; and dead code from rapid churn that now misleads (vestigial pane registry, routing card wired to nothing, persisted view never restored).

Strengths

- Stream coalescer: one store write per animation frame; tail chunks never drop or reorder.
- Battle-hardened xterm layer with sequence-aware transcript replay.
- Enforced engine decoupling via sessionGlue; strong hydration discipline.
- 50+ test suites incl. persistence-migration and 931-line listener tests.
- Code-splitting done right; live PTYs survive navigation.

Problems (ranked)

HIGH Prompt Library "Send to terminal" silently broken via dead layoutStore pane registry

(src/stores/layoutStore.ts)

addPane is never called; sendToTerminal always gets undefined — silent no-op. CLAUDE.md documents the dead path.

HIGH Always-mounted chrome subscribes to the full conversations array per streaming frame

(src/components/layout/LiveSpendChip.tsx)

Spend/agents chips and approval banner re-render at up to ~60 fps; cost aggregation re-runs over every message each frame.

MEDIUM Unmemoized message rows and MarkdownRenderer (src/components/agents/chat/MessageList.tsx)

Mount-once-never-unmount rows re-parse the entire conversation's markdown per frame after scrolling a long transcript.

MEDIUM Persistence fragmented across five mechanisms with hand-rolled duplication

(src/stores/layoutStore.ts)

setProjectPath re-implements workspaceStore persistence inline — a documented drift hazard.

MEDIUM routingStore task-routing UI is a dead end — resolveForTask has zero callers

(src/stores/routingStore.ts)

Users configure routing that has no effect anywhere; wire into launch paths or hide until LM1–LM7 lands.

MEDIUM Persisted selectedView written but never restored — always boots to Welcome (src/lib/bootstrap.ts)

Half the round-trip is dead code; daily users re-navigate every launch.

MEDIUM Monolithic view components with high hook density (src/components/views/GitHubView.tsx)

GitHubView 1204 lines / 23 hooks; WorkspaceCreationModal 1138 / 26; the hardest files to safely modify.

MEDIUM Fragile PTY approval automation and initial-prompt timing heuristics (src/hooks/useTerminalSession.ts)

Approvals write yln on regex TUI detection; non-Claude CLIs get a blind 3 s prompt delay that can swallow the task.

LOW Trailing-edge-only save debounce loses in-flight turns on crash

(src/stores/agentConversationPersistence.ts)

The 500 ms timer never fires during continuous streaming; add a max-interval flush.

LOW ~25 hardcoded "packetade:" literals despite brand.ts; dead files; 4 always-on status pollers; window CustomEvent side-channel (src/lib/brand.ts et al.)

A future rename is again multi-file churn; ServersView dead; 4 IPC polls per 5 s idle; untyped cross-component events.

Recommendations

- Memoize rows + renderer; derived chrome selectors; split conversation metadata from bodies.
- Delete the vestigial pane API; repoint promptStore; fix CLAUDE.md.
- Restore selectedView on boot; centralize storage keys with an eslint ban on raw prefixes.
- Replace the 3000 ms prompt delay with readiness detection; add max-interval save flush.

2.2 Rust Backend — commands, core, supervision, persistence

A genuinely mature Tauri v2 backend. storage.rs implements an atomic write path with a quarantine-first recovery ladder and hermetic tests; api_agent.rs has a correct turn-ownership protocol with cancellation at stream, tool-join, and pending-prompt levels; the supervisor rate-limits restarts and kills process trees on all platforms. Secrets in the OS keyring with careful migration. Production unwraps nearly all test-confined.

Weaknesses are architectural: one state.v1.json fully rewritten (pretty-printed, double-fsynced) on every slice save behind a spin-lock; and a trusted-webview security model — raw ssh argv from the frontend, a cosmetic PTY allowlist including shells, and PermissionMode::Auto running bash/write unprompted while MCP and web tools bypass the gate entirely.

Strengths

- Durable write path provably unable to clobber a good backup; crash-ordering tests.
- Concurrency-correct turn lifecycle with monotonic turn ids and deny-safe draining.
- Backend-owned-field merges prevent stale UI snapshots rewinding attempt state.
- Thoughtful SSH: SHA256 pinning, POSIX quoting, traversal rejection.
- Process hygiene: group kills, orphan reaping, crash fan-out, exit hooks.

Problems (ranked)

HIGH ssh_exec accepts arbitrary argv (and password) from the webview (src-tauri/src/commands/pty.rs:727)
ProxyCommand = local code execution; passwords in IPC. Build argv backend-side from ServerConfig id.

HIGH Default PermissionMode::Auto runs bash/write_file unprompted; gate covers two tool names
(src-tauri/src/commands/api_agent.rs:37)
MCP and web tools bypass the permission gate entirely. Default ask_for_risky; classify MCP/web.

MEDIUM Sync state writers spin on try_lock with 1 ms sleeps, no fairness (src-tauri/src/core/storage.rs:353)
Sync writers can starve indefinitely; bridge to fair async locking.

MEDIUM Monolithic state.v1.json rewritten on every slice save; memory slices unbounded
(src-tauri/src/core/storage.rs:100)
O(total-state) disk churn per UI persist; move to per-slice files or SQLite.

MEDIUM write_pty holds the global manager mutex across a blocking write (src-tauri/src/commands/pty.rs:615)
One wedged child freezes every terminal pane; per-session writer mutexes.

MEDIUM Usage JSONL append not serialized — interleaving can corrupt the cost log
(src-tauri/src/commands/usage.rs:35)
Single pre-joined buffer under a static Mutex.

MEDIUM PTY allowlist cosmetic; env injection unrestricted; in-process history unbounded
(src-tauri/src/commands/pty.rs:25, api_agent.rs:95)
Shells allowlisted and LD_PRELOAD/PATH passthrough; long sessions die on raw provider context errors.

LOW Docs contradict code on GitHub auth; legacy plaintext token path; unknown providers mislabeled as api-claude; timeout masking; 3 home-dir resolvers; 200 hand-registered commands (github.rs, api_agent.rs, lib.rs)
Migrate-and-delete the plaintext token; log unknown providers; adopt tauri-specta to end registration drift.

Recommendations

- Backend-constructed SSH invocations; ask_for_risky default; per-workspace trust toggle.
- Per-slice state files (interim) then SQLite; fair locking; manual restart_sidecar command.

- Context compaction for the in-process loop; typed error enums; tauri-specta.
- `deny(clippy::unwrap_used)` to lock in current discipline.

2.3 Sidecar & Provider Layer

The most architecturally interesting part of PacketADE: a versioned NDJSON stdio protocol (v11 — CLAUDE.md still said v10) connecting the Rust supervisor to a Node sidecar hosting three real providers, plus five in-process Rust providers sharing an OpenAI-compatible core. One event contract keeps the frontend transport-agnostic; the protocol file carries an inline changelog; the MCPH4 trust-snapshot design is genuinely well thought out.

Two systemic weaknesses: the protocol conflates advisory and terminal errors and the supervisor treated every error as session death (FIXED this session); and cancel/edit-approval correctness in the Anthropic provider — `cancel()` aborting the session-lifetime query (FIXED) and `pending_edit` omitting the mandatory `toolUseId` so gated write approvals can never resolve (STILL OPEN). The 14 smoke gates run against the echo provider — exactly the class of bug that shipped has no coverage.

Strengths

- Single api-agent:* contract across two backends; SSH-remote sidecars dropped in cleanly.
- MCPH4 trust snapshots: immutable authority, denial floors, Codex stdio trust proxy.
- Defensive I/O: 8 MB capped reader, UTF-8-safe truncation, SSE edge handling.
- Codex provider shows deep operational knowledge (shims, dual-schema parsing, watchdog).
- Thorough remote preflight; password-auth SSH refused on the protocol channel.

Problems (ranked)

CRITICAL [OPEN] Anthropic `pending_edit` omits `toolUseId` — gated write approvals can never resolve

(`agent-sidecar/src/providers/anthropic.ts`)

The resolver is parked under the id but the event carries none; the SDK stalls on the write tool forever. One-line fix.

HIGH [FIXED] Supervisor treats every error event as session-terminal (`src-tauri/.../agent_sidecar/handler.rs`)

Advisory errors permanently unrouted live sessions and leaked sidecar-side sessions and MCP children.

HIGH [FIXED] Anthropic `cancel()` aborts the long-lived query (v10 contract breach)

(`agent-sidecar/src/providers/anthropic.ts`)

Post-cancel sends pushed into a dead query and spun forever. See bug spotlight, Section 5.

HIGH [FIXED] Codex cumulative token snapshots additively accumulated into Flight cost

(`src-tauri/.../agent_sidecar/handler.rs`)

Sum-of-prefix-sums inflation. See bug spotlight, Section 5.

MEDIUM Cancel done/pump-done race; MCP trust false denials; no manual sidecar restart; Codex retry loses context; `allow_always` not sticky; no ack/nack; echo-only smoke gates (`agent-sidecar/*`)

A stale late done can zero a new turn; `get_dataset` denied for containing "set"; 3 crashes in 60 s costs all subscription chats until app restart; retry can replay a wake trigger; version skew degrades silently.

LOW `StartSessionRequest` god-object with undocumented aliases; v11 changelog drift; echo provider in production list; model-supplied regex in grep (`protocol.ts` / `mod.rs`)

The wire contract is wider than its type; documentation lockstep already broken; catastrophic backtracking can pin the shared event loop.

Recommendations

- Terminal/advisory distinction on the wire + regression test that ownership survives advisory errors.
- Ship `toolUseId` on `pending_edit` (one line) and a `restart_sidecar` command.
- Real-provider lifecycle gates behind env credentials: edit approval, cancel-continue, error-then-send.
- `node:test` runner; generate Rust protocol structs from `protocol.ts`; word-boundary trust heuristics.

2.4 Flights, Worktrees, SSH Remote Execution & the Remote Agents Plan

A genuinely mature one-prompt-N-agents orchestrator: Rust owns attempt lifecycle behind a serialized lock; the frontend re-hydrates from the backend as source of truth. Merge protection, dual-side path-claim collision gating, publish-before-teardown ordering, and Rust-enforced reviewer gates show hard-won correctness. SSH security is notably above average.

The weakest area is failure recovery across process death: startup recovery never touches attempts (zombie running attempts after a crash), Failed transitions force-delete uncommitted agent work, and SSH cleanup on event-driven failures never happens. The Remote Agents plan (Cloudflare relay, PWA-first, desktop-owns-execution) is thorough — and 100% unimplemented; Sprint 0 has not started and two gating decisions remain open.

Strengths

- Backend-authoritative state; terminal attempt states cannot be rewound by stale saves.
- Fail-closed host-key pinning for async launches; askpass-delivered passwords.
- Collision gating re-validated under the state lock; reviewer gate enforced in Rust.
- Cooperative-flight integration branches with conflict pausing and explicit landing.
- Hermetic money-path tests; partial-launch failure rehydration.

Problems (ranked)

CRITICAL [OPEN] No startup recovery for attempts — zombie running attempts after restart/crash

(src-tauri/src/core/orchestrator.rs)

Dead sessions, listeners on events that never fire, leaked worktrees. Resume plumbing exists and is unused.

HIGH [OPEN] Failed/rejected attempts destroy uncommitted agent work (src-tauri/src/core/worktree.rs)

worktree remove --force on every terminal transition; the evidence needed to diagnose failure is deleted.

HIGH [OPEN] SSH remote worktrees orphaned on event-driven failures

(src/stores/asyncAttemptTerminalListeners.ts)

Cleanup is deferred to a frontend call that never comes; the cancel path already demonstrates the backend fix.

MEDIUM Unbounded pkt/* branch and worktree accumulation; no GC (src-tauri/src/core/worktree.rs)

Dozens of stale branches within weeks; .pkt-worktrees/ pollutes git status.

MEDIUM Sequential multi-target provisioning; fingerprint never verified against known_hosts; attempt-level cost never populated; store persistence swallows errors (flight_attempts.rs / execution.rs / flightStore.ts)

4x serial SSH latency; pinning is a boolean; users cannot compare attempts on spend; silent write failures.

MEDIUM Remote Agents plan has zero implementation (dev/remoteargents/06-implementation-plan.md)

No relay, protocol package, PWA, or flag. High-quality plan; "planning package" is the honest status.

LOW Memory brief skipped for SSH targets; draft-PR target follows global repo selection; all-cancelled flights read "paused" (src/stores/asyncFlightStore.ts)

Remote attempts quietly behave worse; PRs can publish against the wrong repo.

Recommendations

- Attempt-aware startup recovery + orphan-worktree sweep — the biggest reliability gap.
- Stash/commit WIP before worktree removal; backend-owned SSH cleanup everywhere.
- Worktree/branch GC; per-attempt cost; first-class attempt diff review.
- Remote Agents Sprint 0 (protocol package, relay + PWA skeletons, flag) and resolve the two open decisions.

2.5 Security

Substantially better than typical for a solo-developed desktop agent tool: keyring everywhere with booleans-only across IPC, defensively engineered remote SSH, strict CSP, monitor-window invoke allowlist, no dangerouslySetInnerHTML. The most serious gap is permission-mode semantics: default "auto" maps to

--dangerously-bypass-approvals-and-sandbox for Codex while mapping to prompting modes elsewhere — three different safety postures behind one label, the most dangerous as default. Several IPC validations are defense-in-depth theater under the honest trusted-webview model; acceptable, but it should be stated.

Strengths

- API keys never cross IPC; keyring failures distinguished from missing credentials.
- Exemplary Unix askpass password path; POSIX-quoted, workspace-confined remote commands.
- Strict CSP; monitor windows restricted to a 5-command read-only allowlist.
- OAuth parsing extracts only expiry/refresh booleans; tokens never leave Rust.

Problems (ranked)

CRITICAL [OPEN] Default "auto" maps to --dangerously-bypass-approvals-and-sandbox for Codex

(agent-sidecar/src/providers/openai-codex.ts)

A stock subscription conversation runs unsandboxed and unapproved; prompt injection becomes silent host code execution.

HIGH Silent TOFU fallback when SSH host fingerprint is unpinned (src-tauri/src/core/execution.rs)

Legacy servers stay in accept-new forever with an invisible warning; badge the state; hard-require pinning for password hosts.

MEDIUM Windows stdin password piping can leak to the remote command; unused npx/node shell grants; unbounded plaintext PTY transcripts; raw ssh argv from the webview (tool_runtime_ssh.rs / capabilities / pty.rs)

The Unix-fixed hazard persists on the primary platform; ACE primitives granted for zero benefit; secrets-at-rest grow forever.

LOW Stale specs-gen.vercel.app CSP entries; MCP rawConfig ships Authorization headers to the webview; caller-supplied fs confinement; auto-install of CLIs on SSH connect; full "repo" scope device flow

(tauri.conf.json / mcp.rs / fs.rs / useServerConnection.ts)

Remove dead allowances; redact secrets pre-IPC; align editor fs with the agent runtime; make remote installs opt-in.

Recommendations

- Permission-mode unification with truthful labels — required before handing the app to anyone else.
- Visible SSH trust state; written threat model in dev/; transcript lifecycle management.
- Security-posture panel showing the effective flags actually passed per session.

2.6 Daily-Driver UX — shell, navigation, keyboard, lifecycle

Well-considered where ambient awareness matters: pinned approval banners, a running-agents-and-spend HUD, PTYs that survive navigation, one status-rollup truth. The unified Composer is the strongest single surface in the app.

What would grind an 8 h/day user down is the keyboard and lifecycle model: five scattered keydown listeners with inconsistent guards, layout-dependent chords, no pane cycling, a nav-only palette missing core destinations, one-click session-annihilating close, forced Welcome on boot, no undo, ~96 console-only failure sites vs one toast consumer, and a flagship launch modal that forgets its targets and ignores Escape. None of it is architectural — the primitives all exist.

Strengths

- PinnedApprovalBanner with one-click jump from any view.
- RunningAgentsChip + LiveSpendChip: a persistent fleet-and-spend HUD.
- One Composer for launch and chat with drafts, attachments, voice, queued sends.
- Flight-launch safety rails; atomic persisted-state writes keep data safe at rest.

Problems (ranked)

CRITICAL [OPEN] Global Ctrl+K/Escape handler fires inside terminals and inputs [UX-08] (src/App.tsx)

No editable-target guard, no xterm custom key handler: readline kill-line pops the palette over your shell.

CRITICAL [OPEN] Close kills all PTYs/agents/attempts with zero confirmation; restart lands on Welcome

[UX-09] (src/components/layout/TitleBar.tsx)

No onCloseRequested guard anywhere; the persisted view is written but never read.

HIGH ~96 console-only failure sites vs one toast consumer; launch modal forgets targets and ignores Escape; palette nav-only and missing Flights/Agents/Cost; no undo anywhere; no workspace/pane keyboard switching [UX-22 / 23-24 / 14 / 25 / 26] (multiple)

Silent data-loss discovery; a re-selection tax on the flagship flow dozens of times a day; "Clear all memory" one native confirm from wiping months of patterns; layout-dependent chords broken on non-US keyboards.

MEDIUM Notification gating inconsistent, OS toasts are dead ends; one-click Bypass-perms toggle; startup blocks on full conversation hydration [UX-29 / 30 / 31] (src/lib/notifications.ts et al.)

Attempt completions require a fully hidden window and bypass prefs; a misclick removes the human gate; cold starts scale with history size.

LOW "New" menu promises sessions it lacks; Settings/Tools naming collision [UX-15 / 40]

(src/components/layout/Toolbar.tsx)

Three related-but-different "tools" concepts coexist.

Recommendations

- Superseded by the reconciled UX Ledger (Chapter 3): the five owner decisions come first, then the P0 quartet, quick wins, and MS-wave sequencing of Section 3.4. Findings above carry their ledger IDs (UX-nn).

2.7 Quality & Infrastructure – tests, gates, CI, release, telemetry

Unusually strong at the unit level, unusually weak at the process level. 165 Vitest files (50 of 54 stores covered), 417 Rust tests concentrated where risk is highest, 13 smoke scripts written from real regressions, a Rust-to-TS schema drift gate, cross-compile bundle assertions. The bundling pipeline is the best code in this review area.

But every gate is voluntary: no CI (explicit decision), no hooks, a full gate long enough to skip under pressure. Coverage is never measured; E2E is 134 lines of web-mode smoke with all IPC mocked; macOS/Linux have never been compiled. No updater, no signing, version triplicated, zero field visibility for crashes.

Strengths

- Real store-level coverage incl. migration fixtures and lifecycle suites.
- Schema drift gate between Rust DTOs and TS bindings.
- SHA256-verified Node fetching; build-failing cross-compile assertions.
- Layered, honest release gating; local crash capture end-to-end.

Problems (ranked)

HIGH No CI of any kind — all gates voluntary at 177k LOC (dev/local-quality-gates.md)

Session logs show multi-week fix loops landing between full gate runs; a dumb push workflow fixes the class.

HIGH macOS/Linux documented but never compiled; no updater, no signing (dev/multi-platform-build.md, dev/updater-setup.md)

The first non-Windows user is the integration test; every release is a manual SmartScreen-flagged reinstall — the updater runbook is complete, only ops work remains.

MEDIUM E2E is mocked-IPC web smoke; coverage never measured; prune-sidecar destroys the dev env; no clippy/fmt gate; zero field crash visibility (playwright.config.ts et al.)

PTY, sidecar handshake, and flight lifecycle have no automated end-to-end coverage; sidecar TS dev silently breaks after every production build.

LOW Bespoke sequential .mjs smoke chain; version triplicated; Playwright keys off a CI env that can never be set (agent-sidecar/test, scripts/release-gate.mjs)

node:test would fix the chain with zero new deps; a bump script beats a detection gate.

Recommendations

- Incremental CI: one 10-minute workflow now, compile matrix later; clippy + rustfmt now.

- Updater first, then signing; release automation as a script that becomes the CI job.
- Measure coverage before writing more tests; one golden-path native smoke test.
- Opt-in crash report submission; persistent frontend error log.

3. The UX Ledger – Three Audits Reconciled

Three UX audits of three different kinds — the 2026-07-29 main-shell/right-panel source audit plus the placebo-settings decision doc (**A29**), this review's daily-driver code critique (**CR**, Section 2.6), and the 2026-07-30 rendered visual audit (**VA**, Chapter 4) — reconciled into one deduped ledger: **43 findings, 5 owner decisions, 6 cross-source merges, 0 resolved**.

Fix-status ground truth. The midway session's 16 confirmed fixes were all store/backend bugs. **None of them touched any finding in this ledger.** The only partial improvement found during 07-30 re-validation: LiveSpendChip's persisted half is now day-bucketed; its live half still sums all hydrated conversations unfiltered (UX-17). Everything else was re-verified still-live, including the Escape-close gap (Modal.tsx:40 defaults closeOnEscape=false; six other modals opt in; LaunchAsyncFlightModal and NewIssueForm do not — verified 2026-07-30).

3.1 Section A – The Five Owner Decisions

The highest-leverage pending calls in the product. All five were requested 2026-07-29, all five are still pending in HANDOFF thread 2, and none of the MS1-MS4 implementation waves can start until they land. Deciding them costs minutes; every week they sit, the findings below stay open.

#	Decision (near-verbatim)	Rec.	Unblocks / resolves
D1	Remove the Workspace-level Agent inspector now, keeping Inspector owned by Agents.	YES	MS1 step 1. Resolves UX-01; removes the worst contributor to UX-02's width contention. (A29 only — CR/VA did not reach this surface.)
D2	Replace independent right panels with one RightDock.	YES	All of MS2. Resolves UX-02, UX-03, UX-10, UX-11; prerequisite for D5. (A29 only.)
D3	Disable unsupported SSH Preview/Diff/Editor actions before adding full remote parity.	YES	MS1 step 2 — disable now, parity later. Resolves UX-04 and UX-13 (flagged independently by both 07-29 docs).
D4	Make one route registry own rail, palette, labels, and hotkeys.	YES	All of MS3. Resolves UX-14 (A29 + CR, identical missing list), UX-15 (the only three-source finding), most of UX-40, the shortcut half of UX-26. (A29 + CR agree in detail; VA corroborates.)
D5	Reconnect the lightweight Editor through the dock or remove its unreachable shell.	DECISION REQUIRED	MS2 step 5. Resolves UX-12. The one decision without a recommended answer — depends on positioning against the sibling PacketCode TUI. (A29 only.)

Adjacent but not one of the five: the settings-decision doc's Decision 6 (remove/disable the placebo settings — UX-05/06/07/19) was approved in principle on 07-29 but never executed; the placebo controls were verified still live on 07-30. It needs execution, not a decision.

3.2 Section B – Unified Finding Table (43)

Severity is the highest claimed by any source (A29 P0=critical, P1=high, P2=medium; CR and VA as stated). Sources: A = 07-29 audit docs, C = code review, V = visual audit. Counts: Critical 9 · High 19 · Medium 13 · Low 2; cross-source merges 6 (UX-07/13/14/15/22/24). The HTML report carries a per-row validity-evidence column (all 43 verified still-open on 07-30; sole partial: UX-17's spend-chip persisted half) — compressed out of this rendition.

ID	Sev	Finding	Src	Blocked by
UX-01	CRITICAL	Workspace mounts a stale global Agent inspector — any selectedConversationId squeezes the CLI workroom with an unrelated Agent rail	A	D1
UX-02	CRITICAL	Right panels (480px Editor + 280px Git + 280–720px Inspector) have no shared width/ownership arbitration; combined widths exceed the 800px minimum window	A	D2
UX-03	CRITICAL	Preview ownership is global and internally inconsistent — previewPaneStore has no conversationId; Hide and Close disagree; later opens update invisibly	A	D2
UX-04	CRITICAL	SSH conversations expose local-only Preview/applied-Review/Undo; Plan's "Hand off to Codex" sets sshTarget: null; aggregate-diff failures become silent zero-line counts	A	D3
UX-05	CRITICAL	MCP provider scope + allowed-tool checkboxes are placebo — mcp_server_start receives only port and allow_writes; Rust builds the full static tool router regardless	A	none (Decision 6 approved, unexecuted)

ID	Sev	Finding	Src	Blocked by
UX-06	CRITICAL	Password-SSH configuration cannot save a password — no set/delete keyring commands, no password field, no host-key/auth/base-path Test	A	none
UX-07	CRITICAL	AI Provider Routing card is unconsumed — resolver has no production call site and selects CLI configs, not the 8 API rows	A, C	none
UX-08	CRITICAL	Global Ctrl+K fires inside terminals and text inputs — no isEditableTarget guard, zero attachCustomKeyEventHandler anywhere; readline kill-line pops the palette over your shell	C	none
UX-09	CRITICAL	Window close kills every PTY/agent/attempt with zero confirmation ; restart force-routes to Welcome with all panes dead, ignoring the selectedView it persists	C	none
UX-10	HIGH	Files advertises a Markdown-Preview path that is not wired — onSelectFile never provided; clicks fall back to copying a path	A	D2
UX-11	HIGH	Plan and Changes each have competing duplicate presentations (chat + Inspector tab + Preview; overlay vs Inspector Diff) with independent collapsed state	A	D2
UX-12	HIGH	Workspace Editor pane is production-unreachable — editorStore.openFile has no production caller; dirty-buffer discard unprotected	A	D5
UX-13	HIGH	Shell + Settings project context lies for SSH Workspaces — Status Strip/useGitInfo poll the stale local projectPath; folder picker can overwrite it; the Workspace Project settings card shows the same wrong local path	A (both docs)	D3
UX-14	HIGH	Navigation metadata duplicated across rail/palette/hotkeys/modules and drifting; command palette omits Agents, Flight Deck, Costs, canonical Dictation ; substring-only matching, no actions/recents/shortcut hints; Dictation has two route identities	A, C	D4
UX-15	HIGH	Creation labels lie throughout the launch path — "+ New" tooltip promises sessions its menu lacks; Fleet "New session" creates an empty Workspace; Ctrl+N makes a Workspace named "New Session"; toolbar "New Flight" opens a modal titled "Launch parallel agents" whose buttons say "Launch agents" — Flight terminology vanishes exactly where a Flight is created	A, C, V	D4
UX-16	HIGH	Git-host capability gating incomplete — Gitea-unsupported checks/AI/draft controls stay visible; repo/host switches don't clear PR detail/diff state	A	none
UX-17	HIGH	Operational indicators report false success — Agent Stop marks idle before backend cancel; Today's-spend live half sums all hydrated conversations; commit-after-review unenforced; Flight Send-to-Monitor drops failures silently	A	none
UX-18	HIGH	Side-chat requests not isolated — closing doesn't cancel the backend stream; unscoped event names let a prior request pollute a new answer	A	none
UX-19	HIGH	"Start right rail collapsed" Agent setting persists but has no production consumer (Inspector owns its own collapsed state)	A	none (Decision 6)
UX-20	HIGH	MCP defaults unsafe on name collisions across scopes — defaults and trust filtering keyed by name, not stable scoped IDs	A	none
UX-21	HIGH	Flight/orchestration settings say "Saved" before persistence — unawaited fire-and-forget writes, swallowed errors, overlapping read/merge/write	A	none
UX-22	HIGH	Error surfacing is broken at both ends: ~96 console.warn/error sites vs one toast consumer means persistence/migration/bridge failures are silent — while the errors that do reach the UI print raw invoke strings verbatim (History body text, Dictation red box) with no styled error component	C, V	none
UX-23	HIGH	New Flight modal forgets picked targets on every launch — useState([]) component state dies with the modal; the flagship flow re-picks agents/branch/base-path dozens of times a day	C	none
UX-24	HIGH	Escape does not close the New Flight or New Issue modals — Modal defaults closeOnEscape to false and neither opts in; muscle-memory dismissal fails in the app's most-used modals	C, V	none
UX-25	HIGH	No undo anywhere; destructive-action confirms use three styles (inline two-step, Modal, native window.confirm); "Clear all memory" is one native confirm from wiping months of patterns	C	none
UX-26	HIGH	Keyboard model gaps — no workspace/pane cycling shortcut at all; view chords keyed on shifted-number glyphs (self-documented broken on non-US layouts); app chords undocumented in-app and unbindable; Settings "Keyboard Shortcuts" card covers only dictation	C	D4 (partial — registry owns shortcut metadata)
UX-27	HIGH	Issues board wraps its sixth column — "Done" lands alone on a second row with a dead right half, at both 1920 and 1280; structural (fixed min-width + wrap), the most broken-looking screen in the app	V	none
UX-28	HIGH	Settings > General notification EVENTS toggles overlap into a fused vertical strip that no longer aligns with its four rows; label-to-control mapping ambiguous; both viewports	V	none
UX-29	MEDIUM	Notification gating inconsistent — attempt/conversation notifications require the window fully hidden and bypass prefs (visible-on-second-monitor users never learn agents finished); OS toasts have no onclick; no in-app notification center	C	none

ID	Sev	Finding	Src	Blocked by
UX-30	MEDIUM	One-click "Bypass perms" toggle flips the workspace's permission gate with only a color change, adjacent to a frequently-clicked toggle — a security-relevant misclick with no confirm	C	none
UX-31	MEDIUM	Startup awaits full conversation-file hydration behind a generic "Loading..." — linearly growing cold-start stall for a daily user	C	none
UX-32	MEDIUM	Left-rail discoverability — icon-only, very low contrast, no labels, no active state on Welcome; first-run users cannot tell the rail is the primary nav	V	none
UX-33	MEDIUM	Agents first-visit onboarding modal auto-opens and its backdrop silently swallows all clicks including left-rail navigation — no response, no visual explanation	V	none
UX-34	MEDIUM	Inconsistent boolean controls — styled toggles vs native checkboxes (Cost Dashboard "On") vs square custom checkboxes (Automation YOLO), sometimes on the same page; explicit-save vs auto-save never signalled	V	none
UX-35	MEDIUM	Three tab idioms in one app — pill tabs (Settings/Dictation), badge-pill tabs (Memory), bare text links (History)	V	none
UX-36	MEDIUM	Empty-state quality uneven — Flight Deck/Memory have designed empty states; History has none (raw "0 prompts"); Workspace shows two competing ones with the onboarding column pushed toward the fold; Agents pairs a detached header band with a floating composer plus redundant new-agent affordances and zero-count filter noise; Dictation shows two mostly-empty panes side by side	V	none
UX-37	MEDIUM	Memory right panel fills only its top ~35% then runs a hard-edged empty column the full viewport height; stacked 10–11px meta rows clutter the top-right corner	V	none
UX-38	MEDIUM	New Flight modal at 720px clips its last row ("Reviewer Gate") mid-checkbox behind the sticky footer with no scroll affordance — looks cut off, not scrollable	V	none
UX-39	MEDIUM	CLI Clients settings: ten near-identical rows restate "not installed" next to nearly invisible status dots (redundant encodings); COMING SOON rows interleaved with installable ones	V	none
UX-40	MEDIUM	A29 P2 bundle — rename GitHub→Git Hosts; Status Strip calls Settings "Tools" while the toolbar's "Tools" means modules (naming collision also flagged by CR); VT→Dictation; "Attach terminal"/"Continue in CLI"/"Open Git ending" mislabels; double-ellipsis Agent header; duplicated Workspace selection; Side Chat can cover the approval banner; missing aria-label/aria-current/menu roles/tablist; no responsive overflow at 800px	A, C	D4 (naming half)
UX-41	MEDIUM	Settings corrections bundle (07-29 decision doc) — wording fixes, SSH delete confirmation/dependency impact, profile/provider validation, notification-permission-denial visibility, ticket-prefix validation, taxonomy rename/merge/delete, live Release Trust status, API-key deep links, reset controls; plus CLI-first preferences, diagnostics doctor, non-secret export/import, documented inheritance	A	none
UX-42	LOW	New Issue modal header style differs from the Launch modal (no icon, lighter weight — two adjacent "create" modals, two treatments); label chips wrap into a ragged mixed-color cloud with unclear selected state	V	none
UX-43	LOW	Layout-balance nits — Settings General masonry hole; Cost Dashboard's lopsided Daily-Guardrail card and accent-green "\$0.00" implying "good" on no data; Automation numeric-caps row reads as a gridline-less table; scope badge easy to miss	V	none

3.3 Section C — Agreements and Conflicts

Where sources agree (strong signal)

- **Naming/label drift in the creation flow — the only three-source agreement.** A29, CR, and VA each independently found that what things are called diverges from what they do — and specifically that "Flight"/"session" terminology dissolves at the exact moments of creation (UX-15/UX-40; largely resolved by Decision 4).
- **Command palette omissions, twice-independently and identically.** A29 P1-9 and CR problem 4 both list Agents, Flight Deck, and Costs as missing — different methods, months apart. Confirms Decision 4's route registry is the right shape of fix.
- **Escape-close, found by code and by pixels.** CR read `closeOnEscape=false`; VA discovered it empirically when the audit script's Escape press was ignored. The cheapest fix in the ledger.
- **Error surfacing has no owned pattern.** CR proved ~96 console-only failure sites vs one toast consumer; VA showed the failures that do reach the UI arrive as raw invoke strings — and identified the in-house model to copy (the Cost Dashboard's dismissible banner).

- **AI Provider Routing placebo re-flagged** independently by the 07-29 settings audit and the midway review (UX-07). **What's good is agreed on too:** dark-theme discipline, the six-mode rail concept, Settings IA, designed empty states, PTY keep-alive, the approval banner/HUD chips. The problem set is drift and wiring, not foundations.

Where sources conflict or one is stale

- **Loud vs silent errors — complementary, not contradictory.** VA shows errors screaming verbatim onto the page; CR shows the app looking fine while data quietly isn't saving. Two halves of the same missing abstraction; one shared error-surface component resolves both (UX-22). VA's specific triggers were E2E-mock artifacts — do not quote those screens' severity without that qualifier.
- **Today's-spend claim is partially stale:** the persisted half is now day-bucketed; the live half remains unfiltered. Quote the narrowed version, not the 07-29 original.
- **VA's coverage boundary is not a clean bill:** it drew no conclusions on the GitHub view, live sessions, live conversations, or data-dense states — precisely where A29's P0 quartet lives. Absence of VA findings on UX-01..04 is a method limit, not disagreement.
- **The settings-decision doc's own status line overstates:** "1-5 done, six-group IA implemented" is true, but Decision 6 remains approved-yet-unexecuted and its P0 placebo findings are all still live. Trust the per-item validation, not the headline status.

3.4 Section D — Recommended Sequencing

1. **Land the five owner decisions** (minutes of owner time, unblocks everything). Four have a recommended "yes"; only D5 needs real deliberation. Simultaneously execute the already-approved Decision 6: remove/disable the placebo controls (UX-05/07/19) and ship or hide password-SSH (UX-06).
2. **The P0 quartet (MS1 — correctness boundaries).** UX-01 remove the unscoped inspector; UX-04 gate SSH local-only actions; UX-03 conversation-scope the Preview; plus MS1's cancellation-acknowledgment and repo-switch items (UX-17/18, UX-16 partial). Alongside: guard Ctrl+K (UX-08) and add close-confirmation + honor the persisted selectedView (UX-09 — two small diffs).
3. **Quick wins (each an afternoon or less, all verified still-open):** Escape-close opt-ins + persist picked targets (UX-24/23); one shared error banner copying the Cost Dashboard's + toast routing (UX-22); the Issues-board grid fix — fit or scroll, never wrap (UX-27); toggle-idiom unification and the fused notification toggles (UX-34/28/35).
4. **MS3 — route registry (D4 executed):** one metadata source generates rail, palette, hotkeys, labels; palette gains missing destinations, actions, fuzzy match; Dictation collapses to one identity; the New menu becomes truthful; GitHub becomes Git Hosts (UX-14/15, UX-26 partial, UX-40 naming half).
5. **MS2 — RightDock (D2 executed):** mutually exclusive right-panel owners, shared resizer, min-center-width collapse; Files-Preview and Changes-Review routed through it; Editor reconnected or removed per D5 (UX-02/03/10/11/12).
6. **MS4 + polish backlog:** ARIA/responsive semantics (UX-40), undo/confirm unification (UX-25), notification center (UX-29), keyboard cycling + rebinding chords (UX-26), onboarding/empty-state pass (UX-32/33/36/37/39), settings corrections (UX-41), packaged cross-platform/SSH visual matrix.

4. Visual Audit (rendered)

The first rendered-pixel pass over the app — screenshots, not code. Captured with e2e/visual-audit.spec.ts (Playwright web mode: Vite dev server + mocked Tauri IPC) at 1920x1080 and 1280x720. The 14 curated captures ship with the HTML report in docs/reports/visual-audit-2026-07-30/; five of the most load-bearing are embedded below.

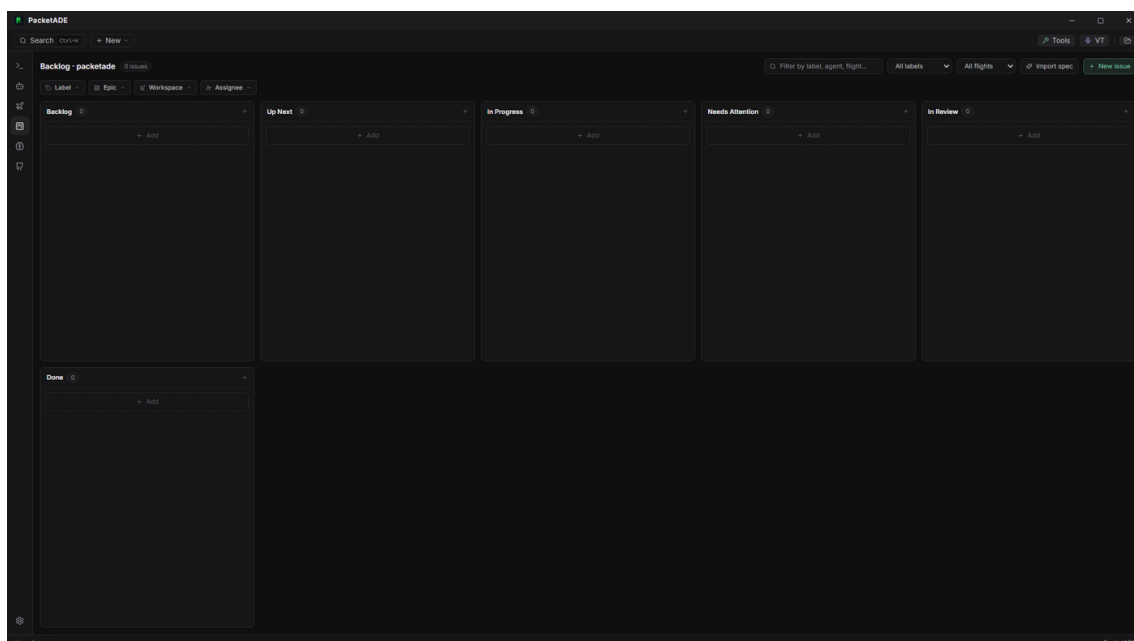
Coverage boundary — what web mode could not audit. All screenshots show true first-run state after `localStorage.clear()`. Not meaningfully auditable, no conclusions drawn: the **GitHub view** (unmocked IPC renders a permanent "Loading..."), **Workspace with live sessions** (PTY/xterm panes never mount in web mode), **Agents with real conversations** (chat stream, tool cards, diff/review UI need a live provider), and **History / Dictation model list / Memory / Cost Dashboard data states** (render empty or show mock-invoke errors — empty-state quality was audited, data-dense layout was not).

Top visual issues (ranked)

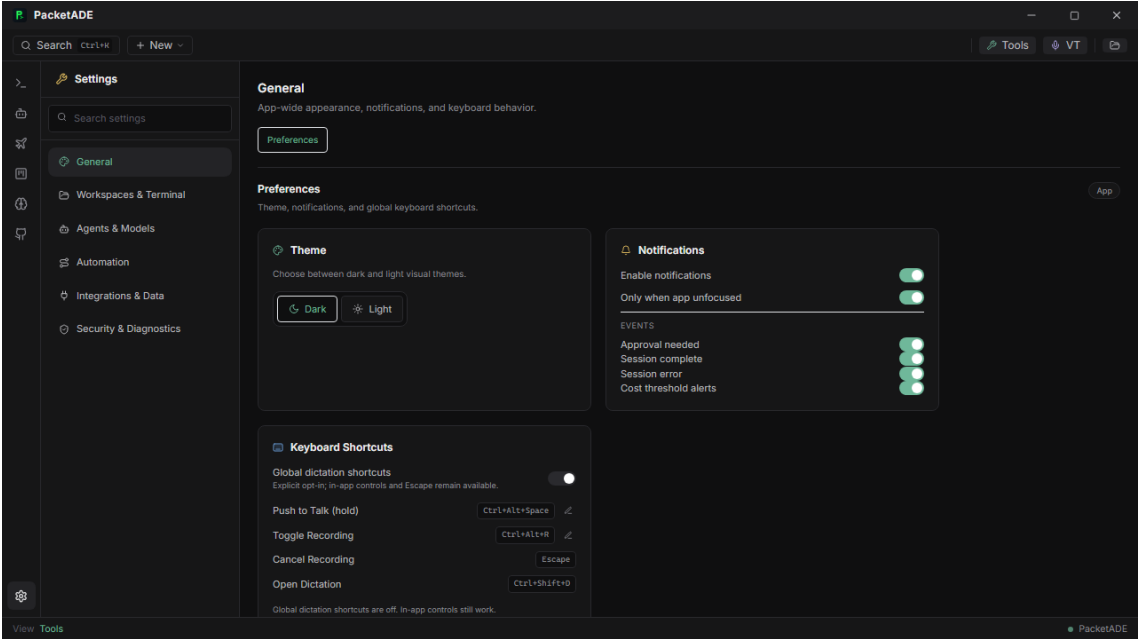
- **Issues board wraps its sixth column** ("Done") onto a lonely second row with a dead right half — at both viewports; structural, the most broken-looking screen in the app [UX-27].
- **Raw backend/invoke error strings rendered verbatim in-page** (History body text, Dictation red box); every error surface should go through a styled component like the Cost Dashboard's banner [UX-22].
- **Settings General notification EVENTS toggles overlap** into a fused strip with ambiguous label-to-control mapping [UX-28].
- **Left-rail discoverability:** icon-only, very low contrast, invisible on first run; the Agents onboarding backdrop silently blocks rail navigation [UX-32/33].
- **Inconsistent boolean controls and three tab idioms** across Settings, Cost Dashboard, Memory, History [UX-34/35].
- **Uneven empty states and dead space** (designed in Flight Deck/Memory, absent in History, competing in Workspace, detached in Agents) [UX-36].
- **Modal keyboard/scroll affordances:** New Flight ignores Escape and clips its last row at 720px; header styling differs between the two create modals; Flight naming drifts through the launch flow [UX-38/24/15/42].

4.1 Selected screenshots (5 of 14)

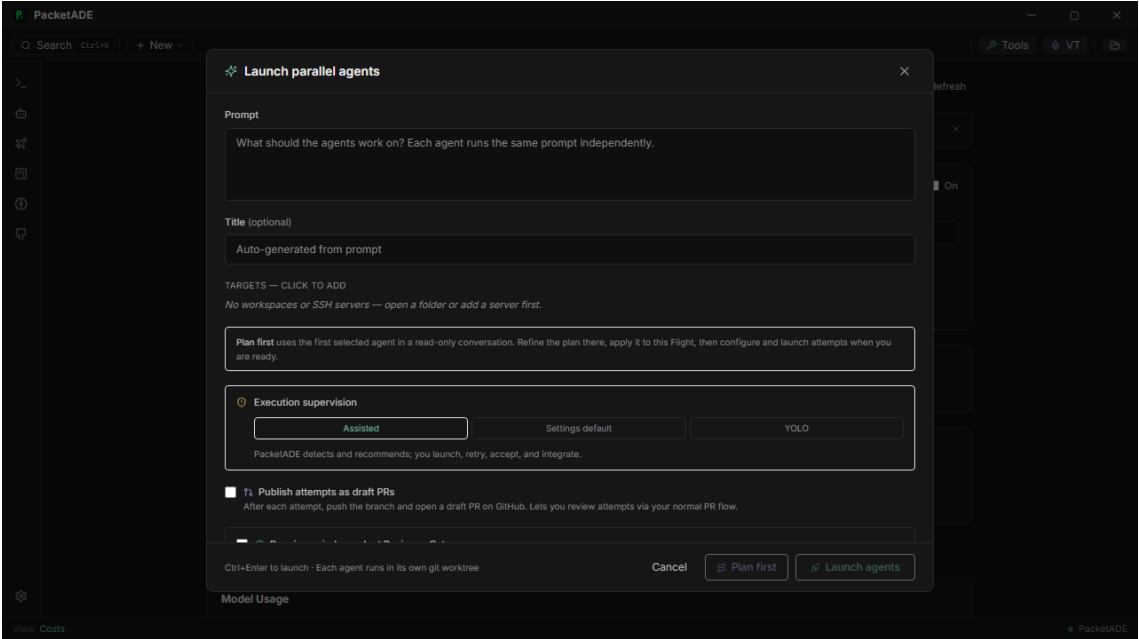
05-issues-board-1920.png — Issues kanban — Done wraps alone onto a second row with a dead right half (high; UX-27; both viewports).



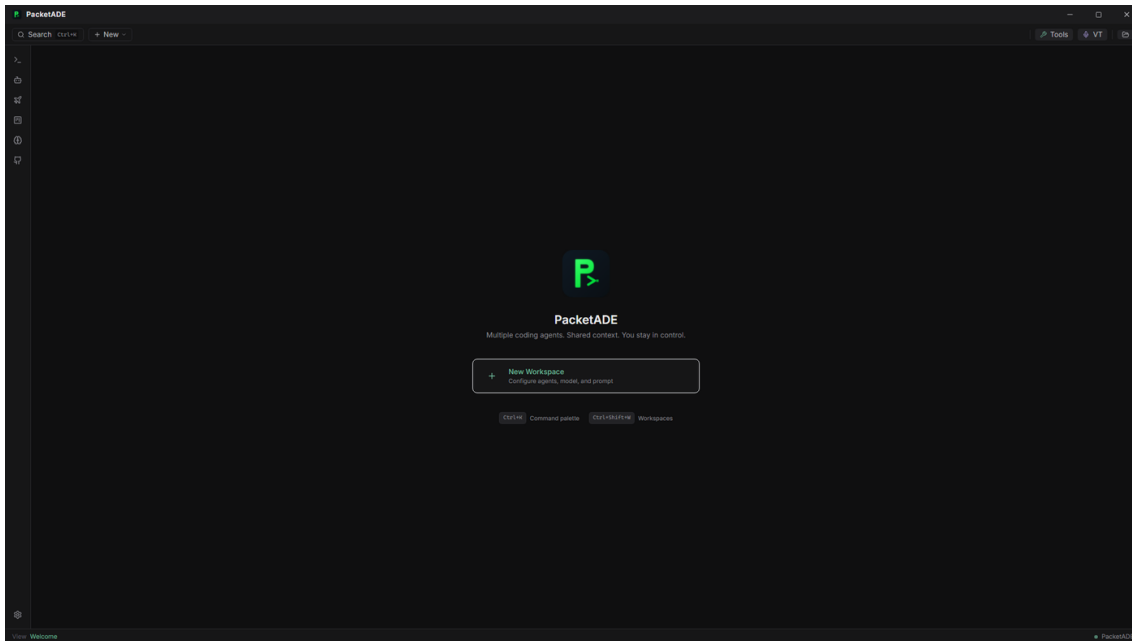
09-settings-general-1280.png — Settings > General at 1280x720 — the four notification EVENTS toggles overlap into a fused strip (high; UX-28).



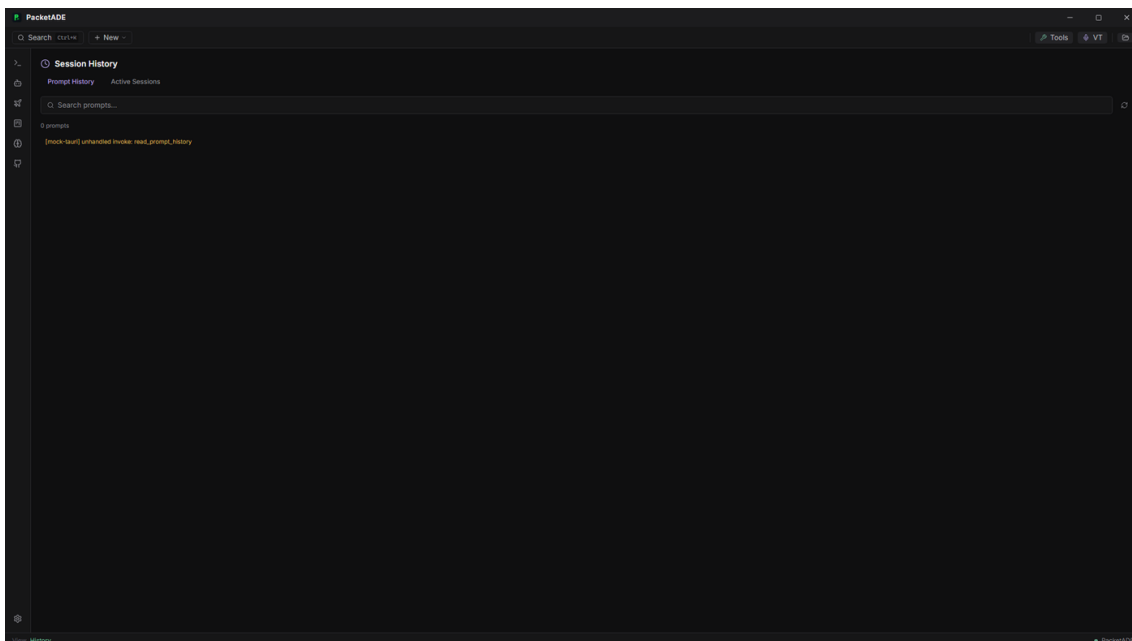
13-modal-new-flight-1280.png — Launch parallel agents modal at 720px — last row clipped mid-checkbox behind the sticky footer, no scroll cue; Escape ignored; "Flight" absent from the title (UX-38/24/15).



01-welcome-1920.png — Welcome — a single small cluster in a ~1880px void; left-rail icons nearly invisible, no labels or active state (UX-32).



07-history-1920.png — Session History — a raw internal invoke error rendered verbatim in the page body; no styled error component (high as rendered; UX-22).



What genuinely looks good: dark-theme token discipline holds everywhere; the six-group Settings IA with pill sub-tabs and scope badges is coherent; concrete previews (commit-trailer live preview, memory-brief code block, shortcut chips) show-not-tell; the command palette surface and toolbar economy are clean; Flight Deck's designed empty state is the model the weaker views should copy.

5. Bug Hunt & Fixes

Six dimension finders, dedup, then adversarial verification with fix designs. All 16 findings survived the verify stage (0 refuted) and all were fixed this session by four fixer agents with disjoint file territories and regression tests. Findings #13 and #15 describe the same TOCTOU defect from two dimensions and were reconciled during fixing — 15 distinct defects repaired.

#	Sev	Title / file	Failure scenario	Fix
1	CRITICAL	Anthropic cancel() aborts the session-lifetime AbortController anthropic.ts:1073	Stop mid-turn, then any follow-up spins forever; all later turns swallowed until app restart.	APPLIED
2	CRITICAL	Codex flight cost over-count: cumulative turn_summary snapshots re-accumulated handler.rs:546	Sum of prefix sums — ~10x cost/token inflation persisted onto Flights.	APPLIED
3	HIGH	packetcode-to-packetade migration runs after store hydration (ESM hoisting) src/main.tsx:6	Upgraders see a reset app; first-session writes permanently overwrite migrated data.	APPLIED
4	HIGH	flight-planner:cost-updated emitted with no frontend listener handler.rs:567	Autonomy budget hard-stop and cost UI run on stale totals; caps overrun by a full attempt's spend.	APPLIED
5	HIGH	UTF-8 char-boundary panic in error_classifier::extract_message error_classifier.rs:186	Long non-ASCII errors panic mid-critical-section; attempts stick in Running forever.	APPLIED
6	HIGH	Reader loop exits on oversized (>8 MB) stdout line, leaving a zombie sidecar supervisor.rs:929	Every sidecar conversation freezes mid-turn with no error, no restart, chip still Ready.	APPLIED
7	HIGH	Every sidecar error event treated as session-terminal handler.rs:600	One transient Codex exit permanently bricks the conversation and leaks the sidecar session.	APPLIED
8	HIGH	Crash-recovery contract broken: "please resend" can never re-create the session agentTaskStore.ts:701	After a sidecar crash every resend fails with "No active session" until app restart.	APPLIED
9	HIGH	Non-conflict merge failure during cooperative integration reported as success worktree.rs:874	Task marked done while nothing merged; Land Flight lands without the accepted work.	APPLIED
10	HIGH	land_local_integration_branch leaves the user's checkout mid-merge worktree.rs:970	Conflicting landings strand MERGE_HEAD and conflict markers in the real repo.	APPLIED
11	HIGH	Pricing table missing default models (claude-opus-4-8, gpt-5, MiniMax-M3) pricing.rs:106	Flight rollups and usage.jsonl record \$0 for the default models; guardrails never trip.	APPLIED
12	MEDIUM	done handler auto-sends queued messages even when the turn was cancelled apiAgentListeners.ts:360	Cancelling an attempt re-sends the queued follow-up 0 ms later.	APPLIED
13	MEDIUM	Concurrent resumeApiConversation installs duplicate listener sets agentTaskStore.ts:431	Chunks applied twice (doubled transcript), spurious failed state, permanent listener leak.	APPLIED
14	MEDIUM	issue-watcher listener in Monitor windows clobbers the issues slice issueStore.ts:774	A stale monitor copy overwrites shared localStorage — comments and links destroyed.	APPLIED
15	MEDIUM	ensureApiAgentListeners TOCTOU (reconciled with #13 as one defect) agentTaskStore.ts:431	Same root cause via the events dimension; single-flight install + resume guard fixes both.	APPLIED
16	MEDIUM	write_file_contents cannot create new files commands/mod.rs:68	Project Rules save always fails on repos missing AGENTS.md or CLAUDE.md.	APPLIED

5.1 Spotlight: the two criticals

Critical #1 — Cancel kills the conversation. On the flagship Anthropic (Subscription) provider, `cancel()` ran the SDK's turn-scoped `interrupt()` and then unconditionally aborted the single session-lifetime `AbortController` owning the long-lived query serving every turn. The pump exited and never restarted while every other layer believed the conversation was alive (protocol v10 makes cancel non-terminal; the registry and Rust both kept the session). The next send pushed into a dead iterable: no chunk, no done, no error — a spinner forever, and every later turn silently swallowed until app restart. **Fix:** cancel is now turn-scoped — parked permission/edit resolvers drained first, `interrupt()` success tracked, session-lifetime abort only as a loud-failing fallback that ends the prompt so follow-ups error instead of hanging.

Critical #2 — Codex flight costs inflated ~10x. The Codex provider emits `turn_summary` on every token-count event with session-cumulative totals (its own comment: "replace, not accumulate"), and the frontend HUD correctly replaces — but the Rust flight rollup priced each snapshot as a per-turn delta and added it. N snapshots $T_1 < \dots < T_N$ accumulated $T_1 + \dots + T_N$ instead of T_N — sum of prefix sums, compounded for sub-agent addresses, persisted into state and shown in the cost chip. **Fix:** provider-aware accounting — for snapshot-semantics providers the handler tracks the last cumulative snapshot per (session, address) and accumulates only the positive component-wise delta with monotonic-reset re-baselining; wire events and frontend unchanged.

Pattern worth naming: 10 of 16 confirmed bugs live in cancel/error/crash/restart seams — precisely the paths deterministic echo-provider gates cannot reach. The highest-leverage test investment is a small set of real-provider lifecycle gates (edit-approval round-trip, cancel-then-continue, error-then-send) behind env-gated credentials.

6. Competitive Landscape & Tech Trends

6.1 The mid-2026 agent-orchestration market

- **Cursor 3** (Apr 2026): Agents Window — up to 8 parallel agents per prompt in isolated worktrees or cloud/remote machines, outputting PRs; computer-use verification; self-hosted cloud agents; Automations; its own fast model (Composer 2).
- **Windsurf -> Devin Desktop** (Jun 2026): Agent Command Center (Kanban of every local + cloud agent), Spaces, multi-agent support via the open Agent Client Protocol (ACP); Devin Cloud from \$20/mo.
- **Conductor**: the category-defining Mac-only app — parallel worktree agents, strong diff/PR flow, free with BYO subscription; used at Linear, Vercel, Notion, Stripe. Limits: macOS-only, no remote execution.
- **Shakeout**: Terragon Labs shut down (Jan 2026, open-sourced); Bloop shut down (Apr 2026) leaving Vibe Kanban community-maintained. Cloud-orchestrator burn rates are fatal; orphaned users are up for grabs.
- **Claude Code**: Agent Teams with fleet guardrails (20-subagent cap, depth 3); Desktop redesigned around parallel agents with per-session worktrees; sandboxed in-app browser; Remote Control (outbound-only sync to mobile).
- **VS Code** is a multi-agent hub (Agent Sessions, native Claude/Codex, /delegate); **Zed + JetBrains** standardized on ACP with an agent registry — "LSP for agents" replacing bespoke PTY scraping.
- **OpenCode / Crush**: BYOK terminal-first engines PacketADE can wrap, not direct competitors.
- **Table stakes** per the 2026 category survey: session visibility, isolation, diff review, workspace organization, resumption. Differentiators: mobile oversight, container sandboxing, SSH/remote execution (only Mux), IDE-level controls.

What PacketADE has that rivals lack

The 8-provider matrix (subscription OAuth + API keys + local models); SSH-remote execution for PTY and agent tools with host-key pinning (only Mux is comparable); Flights above issues/sessions with Kanban, GitHub integration, and per-flight cost rollup (cross-provider cost analytics is absent in every competitor); memory + MCP client/provider hub + prompt library + dictation + deploy in one app; Windows/Linux-native via Tauri.

Strategic implications

- Ship the Remote Agents plan with urgency; the differentiated angle is "your own SSH boxes as the fleet" — self-hosted-first, no cloud bill, reusable in the Anunta enterprise pitch.
- Adopt ACP as a second agent transport mapped onto the api-agent:* contract — one integration buys Devin Local, Junie, Goose, Cline, Auggie.
- Make PRs the first-class output of attempts with inline line-comment diff review.
- Add an agent-verified-outcome step (browser/UI verification pass, "verified" badge).
- Attention routing before full mobile: blocked/needs-review rollup + notifications, then the thin PWA.
- Scheduled/webhook Flight launches as triggers into the existing pipeline — not a planner-FSM revival; optional container isolation for attempts.
- Positioning: "Conductor for Windows/Linux + remotes"; "Devin Desktop without the vendor lock or cloud spend"; target Terragon/Vibe Kanban refugees.

6.2 Technology trends and implications

- **Claude Agent SDK**: nested sub-agents (depth 3), fallback chains, per-agent cost attribution, scoped permissions, subagent streaming; Sonnet 5 with 1M context / 128k output.
- **MCP spec 2026-07-28** (largest revision since launch): stateless HTTP core, Extensions framework, Tasks extension (durable long-running work), MCP Apps (sandboxed-iframe server UIs), auth hardening (iss validation; DCR -> client metadata), simplified elicitation; official Registry targeted Q4 2026.

- **OpenAI:** Agents SDK gained native sandboxing (Python-first; TS lags); Responses API replacing Chat Completions; Codex deprecated --full-auto for permission profiles; app-server JSON-RPC is the documented long-term integration path.
- **Local models:** Ollama 0.24 (MLX, ollama launch); Qwen3-Coder family, Devstral Small 2, Kimi K2.6 are genuinely agent-capable — validating the per-task-class local routing plan (LM1-LM7).
- **Sandboxing is a defined market** (~5 isolation tiers); Anthropic ships srt (Seatbelt/bubblewrap + network proxy); Docker Sandboxes GA'd microVMs; unsandboxed agent tools are becoming unacceptable for enterprise.
- **Platform:** keep xterm.js (WebGL on, Canvas deprecated); Tauri v2 is the safe base (v3 is a draft milestone); the updater plugin is mature and tauri-action produces signed multi-platform installers.

Adopt: MCP 2026-07-28 (+ Tasks mapped onto Flights; MCP Apps spike), registry awareness, Claude SDK upgrades + Sonnet 5, OS-level sandboxing for agent tools, Responses API, Tauri v2 updater now, richer Ollama routing defaults.

Watch/migrate: Codex exec -> app-server; Agents SDK TS sandbox parity. Obsolete: MCP session-stateful assumptions, Chat Completions as primary surface, --full-auto, xterm Canvas renderer, unsandboxed tools, DCR OAuth, manual-only MCP onboarding.

7. Feature Program

7.1 Fix now — open items after this session's repairs

Sev	Item	Domain	Why now
CRITICAL	Codex "auto" -> dangerously-bypass mapping	Security	Default mode silently runs unsandboxed host execution.
CRITICAL	Anthropic pending_edit missing toolUseId	Sidecar	Gated write approvals hang forever; one-line fix.
CRITICAL	Attempt-aware startup recovery	Flights	Crash mid-flight leaves permanent zombie attempts.
CRITICAL	Ctrl+K guards + xterm key handler	UX	The palette pops over your shell on readline kill-line.
CRITICAL	Close-confirmation + selected-view restore	UX	One misclick kills every session; every boot forgets state.
HIGH	Preserve failed-attempt worktrees	Flights	Failures are undiagnosable — evidence force-deleted.
HIGH	Backend-owned SSH cleanup on failure paths	Flights	Remote hosts accumulate orphaned worktrees.
HIGH	restart_sidecar command + chip action	Sidecar	3 crashes in 60 s costs all subscription chats.
HIGH	Backend-constructed ssh argv (retire raw ssh_exec)	Security	ProxyCommand hole; passwords in IPC.
HIGH	Fix/remove Prompt Library send-to-terminal; wire or hide routing card; SSH TOFU badge	Frontend	Settings that silently do nothing erode trust.

7.2 Change — structural

- **Persistence:** per-slice state files toward SQLite; cap memory events; fair locking; failures as toasts.
- **Permission model:** one mode vocabulary across all three runtimes; ask_for_risky default; MCP/web tools gated; an "effective sandbox" field so the UI renders reality.
- **Streaming performance:** memoized rows/renderer, derived chrome selectors, metadata/body split, true transcript virtualization.
- **Contracts over convention:** tauri-specta; protocol structs generated from protocol.ts; typed error enums; terminal/advisory error distinction on the wire.
- **Process:** incremental CI; clippy + rustfmt; coverage measurement; node:test sidecar runner; one native smoke test; single-source version bump.

7.3 Add next — features the reviews rated "needed"

Workflow: view/workspace restore on boot; global search across conversations/transcripts/flights; palette v2 + pane cycling; sticky flight-launch defaults; Escape-to-close; undo via soft-delete; a notification center with click-to-jump; attempt diff review; per-attempt cost.

Platform & trust: auto-updater then signing; compile matrix; worktree GC; known-hosts UI; transcript lifecycle; context compaction for in-process providers; real-provider smoke gates; Remote Agents Sprint 0; written threat model; secret redaction pre-IPC; opt-in crash reports.

Nice-to-have (selected): crash-safe streaming persistence, palette flight relaunch, tile-split layouts, consolidated status poller, light theme, sticky allow_always, sidecar heartbeat, remote sidecar auto-deploy, per-flight repo binding, shortcut cheat sheet, cost sparkline, window-state restore, release automation script.

7.4 Daily-driver blockers — what would make you hate it

- **It fights your hands:** Ctrl+K in terminals; no pane/workspace keyboard switching; chords broken on non-US layouts; Escape ignored by the most-used modals.
- **It forgets:** Welcome-with-dead-panes every launch; flight targets re-picked every time; GitHub re-auth every morning.

- **It loses work silently:** unconfirmed close kills everything; crash loses the in-flight turn; restart zombies attempts; failed attempts delete their work; ~96 silent failure sites; "Clear all memory" one dialog from wiping months.
- **Its trust surfaces lied** [now fixed]: zombie cancels, bricking advisory errors, unusable crash recovery, ~10x Codex cost inflation, \$0 default-model pricing. [Still open]: edit-approval hang, sandbox-less default Codex mode, regex-guessed PTY approvals, silently unpinned legacy SSH hosts.
- **It litters and decays:** pkt/* branches accumulate forever; unbounded plaintext transcripts; whole-state rewrites become visible latency; long in-process chats die on raw context errors.
- **It's hard to keep and share:** manual SmartScreen-flagged updates; never-compiled macOS/Linux; invisible field crashes; auto-installing npm on every SSH host invites bans from controlled environments.

8. The Frontier — Vision Panel & the Top 12

Four lenses generated 38 raw ideas; a synthesis judge merged near-duplicates and ranked the twelve never-built features most worth doing.

8.1 Lens: Fleet Operations & Orchestration

Agent Reputation Ledger — Elo-style win rates from persisted attempt outcomes drive evidence-backed routing.

Flight Recorder — event + git-snapshot journal; scrub any attempt; fork-from-step with an injected correction.

Airspace Control — live cross-flight path/symbol claims; hold, reroute, or sequence before merge conflicts form.

Escalation Auction — capped-budget triage bids compete to win a failed attempt's relaunch.

Fleet Weather — provider health/rate-limit forecasting; ground-stop and divert queued work.

Shakedown Flights — replay your own completed flights as a private eval suite for new models/policies.

Filed Flight Plans — mine retrospectives into reusable templates with track records.

Tournament Mode — bracketed LLM judging over real worktrees with best-of composite assembly.

Convoy Corridors — typed dependency contracts between flights with auto hold/release and rebase.

8.2 Lens: Memory, Knowledge & Context

Flight-to-Runbook Distiller — compile successful flights into executable, parameterized runbooks.

Pitfall Tripwires — learned pitfalls gain triggers; the environment interjects warnings at runtime.

Agent Exit Debriefs — one structured question at session end, deduped into the pattern store.

Closed-Loop Context Compiler — token-budgeted briefs with usage-tracked automatic confidence rerating.

Memory A/B on Parallel Attempts — run the memory counterfactual for free on sibling attempts; measure ROI.

Nightly Consolidation Daemon — merge, decay, and promote patterns into AGENTS.md as reviewable diffs.

Decision Provenance — "why does this line exist" walks blame -> edit -> conversation -> pattern.

Memory Packs & Marketplace — signed, versioned bundles of patterns/runbooks installable like MCP servers.

Cross-Host Memory Mesh — learned patterns sync across the SSH fleet keyed by project.

8.3 Lens: Remote, Mobile & Collaboration

Unified Approval Inbox — risk-ranked triage queue across all hosts and Flights; batch low-risk approvals.

Time-Boxed Autonomy Grants — scoped expiring grants ("writes under src/ for 30 min") with visible countdown.

Agent Presence & Live Handoff — supervision as a transferable seat between desktop and phone.

SSH Fleet Mesh — migrate a running attempt to another capable host when the laptop lid closes.

Morning Brief — signed overnight digest per flight with swipeable merge triage.

Pair Tower — invite a second human with role-scoped approvals and per-human audit.

Dead-Man Escalation Chains — SLA-driven escalation for unanswered approvals; blocked-time metered.

Walkie-Talkie Mode — voice-native supervision with TTS narration and voice-answerable approvals.

Flight Radar — ambient wallboard of every attempt: phase, burn rate, blocked, staleness.

Guest Approval Links — lend one decision via a scoped, expiring capability URL.

8.4 Lens: Trust, Autonomy & Verification

Autonomy Licenses — agents earn expanded YOLO bounds through verifiable track record; breaches demote.

Signed Commit Provenance — attestation trailers linking commit -> attempt -> conversation -> evidence.

Adversarial Verification Lane — a red-team agent must break the diff or concede.

Blast-Radius Budgets — metered write scope per attempt — damage surface, not just spend.

Rollback Contracts — autonomy authorized only after the undo is generated and proven clean.

Taint Counterfactual Replay — re-run tainted turns with evidence redacted; divergence proves steering (moonshot).

Escrowed Claims — "tests pass" held in escrow until a deterministic verifier confirms it.

Policy Flight Recorder — deterministic replay of every autonomy decision; backtest policy changes.

Risk-Priced Verification Router — score each change and route to a proportional review lane.

Honeytoken Tripwires — decoy credentials trip instant hard stops on read/transmit.

8.5 The Top 12 — synthesized, ranked, sequenced

#01 Earned Trust Ledger: Reputation-Weighted Routing & Autonomy Licenses [difficulty: medium]

An append-only local ledger of per-attempt outcomes keyed by (agent x task type x repo area x model), computed from review verdicts, failure categories, cost-per-accepted-diff, and refuted claims — data already persisted but discarded. Drives evidence-backed routing and graduated autonomy tiers; the launch modal shows the receipt.

Sequencing: Build the data-capture layer FIRST — every week of delay is lost training data. Capture + read-only stats panel first; routing and tiers once the record has weight. It is the flywheel hub.

#02 Tournament Mode: Bracketed Judging, Best-of Assembly & Bid-Based Recovery [difficulty: medium]

A reviewer-agent bracket runs pairwise diff comparisons over parallel attempts, ranks a leaderboard, and can assemble a best-of composite in the integration branch; on failure the same machinery runs cheap diagnosis bids. All judgments feed the ledger. No other tool has worktrees + reviewer runtime + integration branch.

Sequencing: After ledger capture and a stable reviewer gate; the highest-visible-payoff first-wave item — it attacks eyeballing N diffs.

#03 Unified Approval Inbox with Leash Slider & Dead-Man Escalation [difficulty: medium]

Every permission/edit/plan approval across all hosts and Flights in one risk-ranked queue; scoped expiring autonomy grants as signed policies with countdown; SLA escalation with blocked-time metered next to cost.

Sequencing: Sequence with the Remote Agents plan — this IS the mobile MVP done right; ship the inbox desktop-first before the relay lands.

#04 Rollback Contracts: Pre-Proven Undo Insurance for Autonomy [difficulty: large]

Autonomous integration only after the revert is generated, dry-run-applied, and proven clean; a permanent one-click "unwind everything this policy did." Verified reversibility as a precondition of autonomy — "YOLO, but fully collateralized."

Sequencing: Build immediately after the auto-integration adapter ships, as its gating precondition; scope v1 to fast-forward-only integration.

#05 Flight Recorder: Time-Travel Replay & Counterfactual Forks [difficulty: large]

Journal every event plus git snapshots at tool boundaries; scrub any attempt with diff and reasoning in sync; fork-from-step into a fresh worktree with an injected correction. The same journal enables policy backtesting.

Sequencing: Start the journaling substrate early (record now, exploit later); defer scrubber and fork UX to a second wave.

#06 Runbook Distiller: Filed Flight Plans from Successful Flights [difficulty: large]

Compile a completed flight into an executable, parameterized template — task graph, routing, prompts, review criteria, budget, verification — offered at launch with its track record attached and versioned by later flights.

Sequencing: Mid-sequence: needs the ledger and a few dozen completed flights; ship distillation-on-completion first.

#07 Risk-Priced Verification: Blast-Radius Budgets + Proportional Review Routing [difficulty: medium]

Per-attempt quotas on files/dirs/LOC/manifests debited live; a risk score (radius, taint, path criticality, license tier) routes each change to a proportional lane, shown contestably on the AttemptTile.

Sequencing: Budgets first (standalone value when YOLO ships), router second; the connective tissue between the trust items.

#08 Closed-Loop Context Compiler with Exit Debriefs & Sleep-Cycle Consolidation [difficulty: large]

Token-budgeted briefs whose items are usage-tracked into automatic confidence rerating; a one-question exit debrief improves intake; a nightly daemon merges, decays, and promotes patterns into AGENTS.md as reviewable diffs. Memory becomes measured, garbage-collected, compounding.

Sequencing: Ship the two small pieces first (debriefs: days; daemon: contained), then usage tracking, then close the loop.

#09 Adversarial Verification Lane (Red-Team Gate) [difficulty: large]

An institutionalized attacker in an isolated worktree: "produce a failing test, crashing input, or injection escalation — or concede." Breaks auto-package as reviewer findings and ding the builder's license.

Sequencing: After Risk-Priced Verification — the adversary only makes economic sense as the high-risk escalation lane.

#10 Fleet Weather: Provider Health Forecasting & Ground Stops [difficulty: medium]

Aggregate rate-limit events, failure categories, and OAuth expiry into a per-provider weather map; ground-stop queued tasks, divert to the healthiest capable agent, hold for quota resets.

Sequencing: After the ledger's routing consumer — weather is the real-time modifier on its historical prior.

#11 Provenance Blame: Signed Agent Authorship + "Why Does This Line Exist" [difficulty: large]

Signed attestation trailers (keyring key, git notes) linking commit -> attempt -> conversation -> evidence, verifiable offline; a blame action walks to the conversation excerpt and the motivating pattern. The compliance story anchoring the enterprise pitch.

Sequencing: Record linkage metadata early; defer signing and blame UI until the trust-provenance loop lands; pull forward if the enterprise pitch goes live.

#12 Runtime Tripwires: Honeytokens + Active Pitfall Guards [difficulty: small]

Decoy credentials tracked by envelope ID trip instant hard stops, pause the Flight, demote the license, and write an incident record; learned pitfalls gain runtime triggers that interject warnings.

Sequencing: Ship the honeytokens half FIRST in absolute sequence — the smallest item, an active intrusion alarm before autonomy expansion raises the stakes. (Cut from the 12: Flight Radar, Walkie-Talkie, Pair Tower, Guest Links, SSH migration, memory mesh, marketplace, convoy corridors, taint replay, presence/handoff.)

9. Actions Taken This Session & Verification

9.1 Fleets run

Research fleet — 14 agents: 7 domain deep dives, 2 web-research briefs, 4-lens vision panel + synthesis judge. **Bug-hunt fleet — 22 agents:** 6 dimension finders -> dedup -> adversarial verify + fix design; 16 confirmed (2 critical, 9 high, 5 medium), 0 refuted; findings #13/#15 reconciled as one defect — 15 distinct defects repaired.

9.2 Code and content shipped (uncommitted at report time)

- **Gemini CLI removed entirely:** 5 files deleted, 40 edited across TS + Rust; persisted "gemini" panes remap to plain terminal; retired agent ids dropped on hydration; the OpenRouter google/gemini-2.5-pro chat model intentionally kept. Both compile gates passed immediately after removal.
- **All 16 confirmed bugs fixed** with regression tests: turn-scoped Anthropic cancel; provider-aware Codex cost deltas; merge machine fails closed (local + SSH) and failed landings abort cleanly; zombie-sidecar tree-kill -> restart; per-turn errors no longer brick sessions; cost event renamed and wired to a listener; migration runs before hydration (ESM side-effect module); crash-recovery resend actually resumes; cancelled turns don't auto-send queues; single-flight listener installs; Monitor windows can't clobber issues; UTF-8 truncation panic fixed; pricing table gained claude-opus-4-8 / gpt-5 family / MiniMax-M3; write_file_contents creates new files with confinement preserved.
- **Statusline** (sibling repo claude-code-tools, committed 84f794f + pushed): Claude Code wiring verified byte-identical; research proved Codex supports only built-in status item ids and OpenCode has none — no fake adapters; a real tmux fallback segment shipped; docs corrected to verified item-id vocabulary.
- **Docs overhaul:** docs/index.html rewritten (fabricated stats, "Missions", Gemini, deleted-planner claims removed; honest status strip); roadmap.html rebuilt on the real R0-R11 tracks; the manual rewritten task-oriented; deferred-work.md re-verified; 14 stale dev/ docs archived; root README/ROADMAP/backlog/HANDOFF/CHANGELOG refreshed preserving the user's uncommitted edits.
- **Marketing:** hero poster, one-pager PDF, social banner, design-philosophy document ("Phosphor Deck").

9.3 Verification gates

Status recorded at report finalization by the orchestrator.

[PASS] pnpm build (tsc + vite, Windows toolchain)

vite built in 33.6 s, zero TS errors.

[PASS] cargo check (Windows toolchain)

only pre-existing ts-rs serde-attribute warnings.

[PASS] pnpm sidecar:build (sidecar tsc)

plus tsc --noEmit clean for app and sidecar.

[PASS] vitest run (src/stores/__tests__ + src/lib/__tests__)

1,260 tests across 166 files (440 Rust tests also pass via cargo test). Two pre-existing load-induced flaky tests were stabilized test-side during the session (timeout budget for cold module imports); every substantive failure found during gating was fixed.

[COMPLETE] Peer-review fleet over the full diff

4 reviewers (Gemini removal, Rust fixes, frontend fixes, docs truth): 3x approve-with-nits, 1x needs-changes; all 12 findings remediated — including one major (order-safe Codex cost delta accounting via per-session sequence stamping) — and re-verified green.

[QUEUED] Windows exe compile

runs immediately after this report is committed; artifacts and hashes will be recorded in HANDOFF.md.

9.4 Notable truths surfaced

- Zero CI, zero coverage measurement, unsigned installers, no updater, no crash telemetry — the quality ladder is discipline-only.

- `PROTOCOL_VERSION` is 11; `CLAUDE.md` said 10 (stale; gitignored/generated — verify against code).
- The "1,261 passing Vitest tests" figure came from a docs-agent count; verify against the vitest gate output before quoting externally.
- Deliberately left for the next loop: Prompt Library send-to-terminal, the no-op routing card, and the never-restored persisted view.

10. Outstanding Audits Ledger

A full validity sweep of every past audit, review, scan, and plan in the repository: **64 documents** inventoried across three territories, **218 open items** extracted, **182 still valid** against the code as of 2026-07-30 — **15 of them critical**. Nothing below is quoted from a document's own (frequently stale) status headline; each row was re-validated with evidence. Some rows bundle several closely-related raw items. This rendition compresses to doc + item + priority; the per-item validity evidence and the superseded/resolved item lists live in the HTML report.

10.1 The 15 critical still-valid items

Thirteen of the fifteen collapse into two owner-decision clusters — the five main-shell decisions (Section 3.1) with their P0 quartet, and the three Remote Agents Sprint-0 decisions — plus the three approved-but-unexecuted placebo-settings removals. Answering the decision clusters dissolves most of this list.

- P0-1: Workspace mounts a stale global Agent inspector — App mounts AgentInspectorPane beside Workspace for any global selectedConversationId (main-shell-navigation-and-right-panel-audit-2026-07-29.md)
- P0-2: Right-side panels (480px Editor + 280px Git + 280-720px Inspector) have no shared width/ownership arbitration; exceed 800px minimum window (main-shell-navigation-and-right-panel-audit-2026-07-29.md)
- P0-3: Preview ownership is global and internally inconsistent — previewPaneStore has no conversationId; Hide/Close disagree (main-shell-navigation-and-right-panel-audit-2026-07-29.md)
- P0-4: SSH conversations expose local-only Preview/applied-Review/Undo operations; Plan's Hand off to Codex sets sshTarget: null; aggregate diff failures become zero-line counts (main-shell-navigation-and-right-panel-audit-2026-07-29.md)
- Owner decision 1 (verbatim-ish): 'Remove the Workspace-level Agent inspector now, keeping Inspector owned by Agents. Recommended: yes.' — UNDECIDED (main-shell-navigation-and-right-panel-audit-2026-07-29.md)
- Owner decision 2: 'Replace independent right panels with one RightDock. Recommended: yes.' — UNDECIDED (main-shell-navigation-and-right-panel-audit-2026-07-29.md)
- Owner decision 3: 'Disable unsupported SSH Preview/Diff/Editor actions before adding full remote parity. Recommended: yes.' — UNDECIDED (main-shell-navigation-and-right-panel-audit-2026-07-29.md)
- P0: MCP provider scope + allowed-tool checkboxes are placebo — mcp_server_start receives only port and allow_writes; Rust builds the full static tool router (workspace-agent-settings-decision-2026-07-29.md)
- P0: password SSH configuration cannot save a password — no set/delete keyring commands, no password field, no host-key/auth/base-path Test (workspace-agent-settings-decision-2026-07-29.md)
- P0: AI Provider Routing is unconsumed — resolver has no production call site; selects CLI agent configs not the 8 API rows (workspace-agent-settings-decision-2026-07-29.md)
- Thread 2: step through the five main-shell/right-dock owner decisions before any implementation; then run only approved MS1-MS4 slices (HANDOFF.md)
- Sprint-0 (a): Auth provider build-vs-buy for v1 (passkey/magic-link on Workers vs Clerk/Auth0/Stytch/Supabase vs Cloudflare Access for internal beta) — owner: Security/Auth agent (09-open-decisions.md)
- Sprint-0 (b): Payload-encryption launch gate (plaintext internal-only vs required-before-external-private-beta) — owner: project owner + Security/Auth agent (09-open-decisions.md)
- Sprint-0 (c): Code location (in-repo remoteagents/ vs apps/ vs separate repo) — owner: implementation lead (09-open-decisions.md)
- Unchecked tail item: hold the Remote Agents Sprint-0 kickoff (lock three decisions, scaffold behind compile-off flag) (sprints-2026-06-15.md)

10.2 Dev root – active planning docs (dev/*.md, backlog, ROADMAP, HANDOFF, docs/deferred-work)

Doc	Item (still-valid, re-verified 07-30)	Priority
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P0-1: Workspace mounts a stale global Agent inspector — App mounts AgentInspectorPane beside Workspace for any global selectedConversationId	CRITICAL
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P0-2: Right-side panels (480px Editor + 280px Git + 280-720px Inspector) have no shared width/ownership arbitration; exceed 800px minimum window	CRITICAL
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P0-3: Preview ownership is global and internally inconsistent — previewPaneStore has no conversationId; Hide/Close disagree	CRITICAL
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P0-4: SSH conversations expose local-only Preview/applied-Review/Undo operations; Plan's Hand off to Codex sets sshTarget: null; aggregate diff failures become zero-line counts	CRITICAL
main-shell-navigation-and-right-panel-audit-2026-07-29.md	Owner decision 1 (verbatim-ish): 'Remove the Workspace-level Agent inspector now, keeping Inspector owned by Agents. Recommended: yes.' — UNDECIDED	CRITICAL
main-shell-navigation-and-right-panel-audit-2026-07-29.md	Owner decision 2: 'Replace independent right panels with one RightDock. Recommended: yes.' — UNDECIDED	CRITICAL
main-shell-navigation-and-right-panel-audit-2026-07-29.md	Owner decision 3: 'Disable unsupported SSH Preview/Diff/Editor actions before adding full remote parity. Recommended: yes.' — UNDECIDED	CRITICAL
workspace-agent-settings-decision-2026-07-29.md	P0: MCP provider scope + allowed-tool checkboxes are placebo — mcp_server_start receives only port and allow_writes; Rust builds the full static tool router	CRITICAL
workspace-agent-settings-decision-2026-07-29.md	P0: password SSH configuration cannot save a password — no set/delete keyring commands, no password field, no host-key/auth/base-path Test	CRITICAL
workspace-agent-settings-decision-2026-07-29.md	P0: AI Provider Routing is unconsumed — resolver has no production call site; selects CLI agent configs not the 8 API rows	CRITICAL
HANDOFF.md	Thread 2: step through the five main-shell/right-dock owner decisions before any implementation; then run only approved MS1-MS4 slices	CRITICAL
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-5: Files advertises a Markdown-Preview path that is not wired (onSelectFile not provided by AgentInspectorPane)	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-6: Plan and Changes each have competing duplicate presentations (chat, Inspector tab, Preview; overlay vs Inspector Diff)	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-7: Workspace Editor panel is production-unreachable — editorStore.openFile has no production caller; dirty-buffer discard unprotected	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-8: Shell project context lies for SSH Workspaces — Status Strip/useGitInfo keep polling stale local projectPath; folder picker can overwrite it	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-9: Navigation metadata duplicated across Left Rail, Status Strip, command palette, hotkeys, modules; palette omits Agents/Flight Deck/Costs/canonical Dictation; Dictation has two route identities	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-10: Creation labels lie — global New menu lacks session; Fleet 'New session' creates an empty Workspace; Ctrl+N creates Workspace named 'New Session'	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-11: Git-host capability gating incomplete — Gitea-unsupported checks/AI/draft controls remain visible; repo/host switches don't clear PR detail/diff state	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-12: Operational indicators report false success — Agent Stop marks idle before backend cancel; Today's-spend live half sums all hydrated conversations; commit-after-review unenforced; Flight Send-to-Monitor drops failures silently	HIGH

Doc	Item (still-valid, re-verified 07-30)	Priority
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P1-13: Side chat requests not isolated — closing doesn't cancel backend stream; unscoped event names let a prior request pollute a new answer	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	Owner decision 4: 'Make one route registry own rail, palette, labels, and hotkeys. Recommended: yes.' — UNDECIDED	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	Owner decision 5: 'Reconnect the lightweight Editor through the dock or remove its unreachable shell. Decision required after reviewing PacketCode/editor positioning.' — UNDECIDED	HIGH
workspace-agent-settings-decision-2026-07-29.md	P1: 'Start right rail collapsed' Agent setting persists but has no production consumer (Inspector owns its own collapsed state)	HIGH
workspace-agent-settings-decision-2026-07-29.md	P1: MCP defaults unsafe on name collisions across scopes — agent defaults and trust filtering persist/filter by name, not stable scoped IDs	HIGH
workspace-agent-settings-decision-2026-07-29.md	P1: remote Workspace Project card shows global local layoutStore.projectPath, wrong for SSH Workspaces	HIGH
workspace-agent-settings-decision-2026-07-29.md	P1: Flight/orchestration settings say Saved before persistence — unawaited fire-and-forget backend writes, swallowed errors, overlapping read/merge/write	HIGH
local-model-routing.md	LM1 (P1, blocks the rest): Ollama fundamentals — native /api/chat with num_ctx + keep_alive in core/llm_ollama.rs, /api/show tool-capability probe with picker gating, visible over-context warning instead of silent truncation	HIGH
backlog.md	Dictation P1: live microphone acceptance matrix (default/USB/BT, 44.1/48kHz, fast-PTT, cancel, disconnect, history, clipboard, external-app) — host has no active capture endpoint	HIGH
backlog.md	Remote Agents: BLOCKED on three Sprint-0 decisions (auth provider, payload-encryption timing, code location) — open since 2026-06-15; no remoteagents/ code until resolved. Then P1s: Packet Cloud relay MVP, account sign-in + device trust, PWA conversation shell, secure remote command envelope, E2E...	HIGH
backlog.md	Settings P1s: enforce-or-remove placebo controls (AI Provider Routing, rail collapse, MCP scope/tool checkboxes); complete-or-hide SSH password config; authoritative safety-setting saves. P2s: scoped MCP server IDs + real active-Workspace project identity + provider-aware profile validation; CLI...	HIGH
backlog.md	Main-shell audit section: P0 inspector scope / one right-dock / Preview+SSH boundaries; P1 nav registry, shell context + Gitea gating, honest operational controls; P2 labels/accessibility — all pending the five owner decisions	HIGH
backlog.md	Local model routing LM1-LM7 (P1 LM1 blocks section)	HIGH
backlog.md	Midway review P2: triage docs/reports/midway-review-2026-07-30.html recommendations into concrete backlog items — its 'Fix now' list (now Section 7.1) (Codex auto→dangerously-bypass mapping, Anthropic pending_edit toolUsed, attempt-aware startup recovery, Ctrl+K/global-shortcut guards, close-con...	HIGH
ROADMAP.md	R11/D14: main shell + right dock — review complete, five owner decisions pending, then MS1-MS4	HIGH
ROADMAP.md	R0/D1: Remote Agents paused at three Sprint-0 decisions (auth provider, E2EE timing, code location)	HIGH
ROADMAP.md	D13: Settings authority cleanup — remove/enforce placebo controls, authoritative safety saves, SSH/MCP/provider correctness before new preferences	HIGH
HANDOFF.md	Settings authority cleanup list (MCP scope/tools, SSH password, AI Routing, rail collapse, scoped MCP IDs, project identity, awaited safety saves)	HIGH
HANDOFF.md	Remote Agents: three Sprint-0 decisions must be resolved by the owner before any remoteagents/ code	HIGH
main-shell-navigation-and-right-panel-audit-2026-07-29.md	P2 bundle: rename GitHub→Git Hosts, Tools/Settings/VT/Attach-terminal/Continue-in-CLI/Open-Git-ending label fixes; two-ellipsis Agent header; duplicated Workspace selection; Side Chat covers approval banner; ARIA (aria-label/aria-current/menu roles/tablist); responsive overflow at 800px	MEDIUM
workspace-agent-settings-decision-2026-07-29.md	Corrections list: PacketAgent 'Test connection'→'Test endpoint health' wording; SSH server delete confirmation/dependency impact; profile model/tool validation vs provider capability; provider key validity tests; notification-permission-denial visibility; issue ticket-prefix validation + taxonomy...	MEDIUM
workspace-agent-settings-decision-2026-07-29.md	Settings product-quality additions: terminal/CLI preferences, provider/CLI/SSH diagnostics doctor, non-secret export/import, documented inheritance model	MEDIUM

Doc	Item (still-valid, re-verified 07-30)	Priority
workspace-agent-settings-decision-2026-07-29.md	Reliability tests: runtime-consumption assertions, SSH secret lifecycle, effective MCP policy, orchestration save failure, Settings deep-link integration tests	MEDIUM
workspace-agents-wa3-handoff-evidence.md	Gate 1: manually exercise local and SSH handoffs in the packaged app	MEDIUM
workspace-agents-wa3-handoff-evidence.md	Gate 2: exercise PacketCode paste/launch with a published PacketCode build	MEDIUM
workspace-agents-wa3-handoff-evidence.md	Gate 3: run the configured live PacketAgent close/restart/evidence proof	MEDIUM
workspace-agents-completion-audit-2026-07-29.md	SSH Workspace/handoff runtime proof — Unavailable (no configured ServerConfig)	MEDIUM
workspace-agents-completion-audit-2026-07-29.md	Published PacketCode paste/launch proof — Unavailable (no installed executable)	MEDIUM
workspace-agents-completion-audit-2026-07-29.md	Credentialed PacketAgent close/restart interop — Unavailable (no configured endpoint/token/listener)	MEDIUM
workspace-agents-wa0-route-contract.md	WA1B agent-first surface: 'Implemented; manual UX sign-off open' — manual product/dogfood review never recorded	MEDIUM
github-pane-v9-loop.md	GP7 remaining gate: packaged GitHub + Gitea live-host proof (create/adapt/update/pull/conflict, hidden-window pause, restart, revoked-auth recovery)	MEDIUM
issue-flight-mirror-design.md	Packaged GitHub + Gitea matrix: create/adapt/update/pull/conflict, hidden-window pause, restart, revoked-auth recovery	MEDIUM
ssh-remote-loop.md	S11 (P2): live Codex-over-SSH smoke — one real remote host with remote Codex auth + installed sidecar, per sidecar-over-ssh-verification.md step 12	MEDIUM
sidecar-over-ssh-verification.md	Manual step 12: live Codex-over-SSH smoke (remote ~/.codex/auth.json + remote sidecar; multi-turn resume on one host) — the only remaining Codex-over-SSH verification	MEDIUM
sidecar-over-ssh-verification.md	Full manual SSH parity checklist steps 1-11 (Anthropic subscription over SSH) have no recorded live pass either — same missing-host gate	MEDIUM
send-to-monitor-plan.md	Packaged manual matrix: multi-display move, close-with-main-process per platform, packaged WebView-to-Rust denial integration proof, stale-state on deleted entity	MEDIUM
local-model-routing.md	LM2 (P2): custom OpenAI-compatible endpoint provider row (user base_url, optional key, manual model list) covering vLLM/LM Studio/LiteLLM/self-hosted gateways	MEDIUM
local-model-routing.md	LM3 (P2, load-bearing): unified auxiliary LLM entry point core/aux_llm.rs with TaskClass — ~15 call sites across 3 incompatible mechanisms (4 get_provider, 3 sidecar claude-oauth one-shots, 4 claude CLI shell-outs)	MEDIUM
local-model-routing.md	LM4 (P2): migrate mechanism-3 shell-out sites (memory.rs, insights.rs, spec.rs, github.rs:1577) off run_claude — removes hard claude-on-PATH dependency, adds token accounting	MEDIUM
local-model-routing.md	LM5 (P2): migrate mechanism-2 sidecar sites (code_quality.rs, github.rs PR commands, issues.rs spec import) with claude-oauth kept selectable so subscription-funded default survives	MEDIUM
competitors.md	Gap: mobile/remote/voice steering (Omnara/AgentsRoom/Cursor own it; PacketADE desktop-only)	MEDIUM
updater-setup.md	Auto-updater deferred pending: signing keypair (offline), hosted signed latest.json + release pipeline, and an update-prompt UI surface	MEDIUM
beta-distribution-trust-runbook.md	Trust gates pending external credentials: Windows signing cert (SmartScreen reputation), macOS codesign/notarization, updater signing key — strict release gate cannot pass until acquired	MEDIUM
deferred-work.md	1. Codex interactive per-command approvals — modeToCodexFlags pins approval_policy=never (stdin closed for codex exec); revisit if codex 0.142+ exposes an approval channel / app-server mode (ROADMAP Later A6)	MEDIUM
deferred-work.md	3. Per-conversation costCapUsd mid-run cancel — Flight-level maxTotalCost gates autonomy but nothing halts a runaway single run (more relevant with MAX_TOOL_ITERATIONS=150)	MEDIUM
deferred-work.md	Validation TODO: autonomy harness (agent-harness.ts) never verified on a real multi-step task — highest-value babysit	MEDIUM
backlog.md	Dictation P2: packaged macOS/Linux matrix (mic/accessibility prompts, ALSA/PipeWire, X11/Wayland)	MEDIUM
backlog.md	SSH: S11 live Codex-over-SSH smoke (P2), S9 Windows-OpenSSH hosts (P3), S10 SFTP/streamed cap + port-forward (P3) — all environment-gated	MEDIUM

Doc	Item (still-valid, re-verified 07-30)	Priority
backlog.md	Platform P2: auto-updater (keypair, hosted signed latest.json, prompt UI); macOS signing+notarization; Windows Authenticode. P3: Snap/Flatpak; cross-compile (won't-fix until matrix demands)	MEDIUM
backlog.md	Flight Deck residual gates on shipped loops: RG8 reviewer-gate, CG9 cooperative-graph, AP9 adversarial YOLO, CI9 inbox delivery — each a release-like local/SSH/manual smoke	MEDIUM
backlog.md	PacketAgent handoff still open: live W9 close/restart/reconnect gate with credentials; direct approval-response route (PacketAgent publishes attention events but no approval-response endpoint); task/conversation source builders; richer attention/cost projection; packaged evidence/artifact return...	MEDIUM
backlog.md	PacketCode: PC5 published signed artifacts + clean-machine install/upgrade/rollback; PC9 hardening queue — PCH3 versioned workflow verifier/retry (P2), PCH4 abandoned-job reconcile/resubmit (P2), PCH5 Streamable-HTTP MCP trust contract (P3), PCH6-PCH8 external release gates; PC10 PacketAgent comp...	MEDIUM
backlog.md	GitHub pane P2: Issue■Flight live host proof (packaged GitHub+Gitea matrix)	MEDIUM
backlog.md	Memory Hub P2: MH8/MH9 packaged interoperability proof (real editor/watch-storm/partial-write/rename/restart + empty/large/dirty/gitignored project smoke)	MEDIUM
backlog.md	MCP Hub P2: MCPH3/MCPH8 — real Codex CLI + local/SSH crash/reload/version-skew, offline install/removal, trust downgrade/reconnect, remote-profile parity, packaged provider smoke	MEDIUM
backlog.md	Trust/provenance P2: TP8 live local/SSH all-provider transport parity, MCP remote, restart, YOLO, packaged visual/manual smoke (denial floors must not weaken)	MEDIUM
backlog.md	Monitor P2: packaged/manual multi-display matrix + close-with-main + packaged denial proof	MEDIUM
backlog.md	Workspace/Agents: WA1 manual product sign-off; WA2 SSH + published-PacketCode proof; WA3 manual local/SSH + external-runtime proof; P2 detachable-window single-writer prerequisite	MEDIUM
ROADMAP.md	R2/D5: distribution blocked on Windows+macOS signing certificates; then signing config + Tauri updater wiring	MEDIUM
ROADMAP.md	R3/D6: Flight Deck supervision — packaged local/SSH/manual smoke matrices (RG8/CG9/CI9/AP9)	MEDIUM
ROADMAP.md	R4/D7: PacketCode release proof — signed artifacts, clean-machine install/upgrade/rollback, cross-product smoke	MEDIUM
ROADMAP.md	R5: PacketAgent live cross-repo close/restart proof	MEDIUM
ROADMAP.md	R6/R7/R8/R9 + D8-D11: Memory Hub, Dictation (DV17 + mic matrix), Trust/provenance TP8, MCP Hub MCPH3/MCPH8 — all environment-gated packaged proofs	MEDIUM
ROADMAP.md	D4: SSH parity — live Codex-over-SSH smoke when a configured remote host exists	MEDIUM
HANDOFF.md	Environment/release gates list: signing credentials, updater signing + hosted latest.json, dictation mic matrices, local/SSH provider/MCP/GitHub/Gitea packaged proof, PacketCode clean-machine proof, PacketAgent live proof, packaged Flight matrices, external-editor/Memory watch interop	MEDIUM
main-shell-navigation-and-right-panel-audit-2026-07-29.md	Evidence-limits gate: packaged proof still required for cross-platform WebView visuals, live SSH error presentation, real GitHub/Gitea accounts, native keyboard/screen-reader behavior	LOW
workspace-agent-settings-decision-2026-07-29.md	Evidence not established: telemetry for session placement; live validation of Settings against configured external accounts/servers	LOW
workspace-agents-wa3-handoff-evidence.md	Detachable interactive Agent windows out of scope until WA5 single-writer contract	LOW
workspace-agents-wa4-dogfood-gate.md	WA5 detachable interactive Agents window remains a later item requiring a single-writer state contract	LOW
workspace-agents-restructuring-goal.md	WA5 — detachable-window prerequisite: single-writer state contract, revisioned multi-window persistence/approval tests, read-only projection first	LOW
github-pane-v9-loop.md	Deferred (not in loop): full Actions/CI dashboard beyond GP6's read view (re-run, log streaming)	LOW
github-pane-v9-loop.md	Deferred: Gists view (split out of GP6 scope)	LOW
issue-flight-mirror-design.md	Deferred to later phase: assignees and comment-thread mirroring (cross-host identity matching)	LOW
issue-flight-mirror-design.md	Flight tasks do not own labels — v1 preserves host labels; a local task-label model is a future decision	LOW
issue-flight-mirror-design.md	Webhooks (vs 60s polling) deferred — needs a public callback endpoint	LOW

Doc	Item (still-valid, re-verified 07-30)	Priority
ssh-remote-loop.md	S9 (P3): Windows-OpenSSH remote hosts — POSIX shell scaffolding ([-e], dirname, heredocs, readlink) breaks on cmd/PowerShell remotes; needs OS-detection probe + unit-tested Windows command builders	LOW
ssh-remote-loop.md	S10 (P3): SFTP/streamed transfer to lift the 2 MB remote file cap (MAX_FILE_SIZE in tool_runtime_ssh.rs), then port-forward — needs a live remote to verify chunk/reassembly correctness	LOW
ssh-remote-loop.md	Deferred beyond loop: full Windows-remote parity (PTY quirks, path translation) and interactive port-forward UX	LOW
send-to-monitor-plan.md	Sprint 4 (later): read-only Approval Monitor and Cost Monitor routes	LOW
send-to-monitor-plan.md	Sprint 5 (later): multi-window expansion — dynamic monitor-agent-<id> labels, saved bounds, duplicate-route focus, restore preferences	LOW
send-to-monitor-plan.md	Sprint 6 (later, blocked): Workspace/Terminal Monitor — requires PTY session-attachment model (useTerminalSession kills PTY on unmount)	LOW
send-to-monitor-plan.md	Future elevated approval-capable Monitor lease (short-lived, toolId-bound) if approval-from-monitor is ever wanted	LOW
local-model-routing.md	LM6 (P3): new modelRoutingStore + settings slice mapping task class → provider/model (NOT orchestrationSettingsStore)	LOW
local-model-routing.md	LM7 (P3): local-vs-metered spend split in CostDashboardView + local/SSH/packaged gates — the feature's proof	LOW
local-model-routing.md	Open question: which task classes default local vs frontier (leaning frontier-default)	LOW
local-model-routing.md	Open question: are subagent/custom-agent tools auxiliary or agentic (currently Haiku but run tool loops)	LOW
local-model-routing.md	Open question: fallback policy when local endpoint is down mid-task — fail vs silent frontier escalation (cost-surprise risk)	LOW
competitors.md	Gap: native-Mac polish and provider row count (Superset 12+, AgentsRoom 8 CLIs) vs PacketADE's 8 rows	LOW
competitors.md	Implication: durable moats to keep investing in — provider/auth abstraction, dual transport, memory, MCP breadth (lanes the field under-invests in)	LOW
deferred-work.md	2. Per-launch MCP multi-select in the Composer (Settings-level default exists; mid-session hot-swap unsupported since v11 frozen trust)	LOW
deferred-work.md	4. Cross-provider reasoningEffort/thinking-budget launch control (Anthropic budget hardcoded 8000 in api_agent.rs)	LOW
deferred-work.md	5. Full agentTaskStore module split (now ~1,520 lines; 50+ importers, incremental slice extraction)	LOW
deferred-work.md	6. Diff viewer extra controls: word-wrap, unified/split toggle, copy diff/file, expand-context	LOW
deferred-work.md	Validation TODO: PTY orphan reaper live verification (close pane, crash/force-quit sweep on next launch)	LOW
deferred-work.md	Validation TODO: Codex streaming feel — codex exec emits complete blocks; investigate delta mode only if it matters	LOW
backlog.md	Dictation P3: Parakeet/Whisper-acceleration benchmark only after packaged CPU-path measurements	LOW
backlog.md	Flight Deck P3: structured per-target partial multi-target launch wire result (today recovered by rehydrate+diff)	LOW
backlog.md	Flight Deck P3: migrate/prune orphaned Planner data — legacy planner_* fields/journals kept read-compatible; define eager migration + release-age gate before deletion	LOW
backlog.md	Mission→Flight compat removal: 2 Rust serde aliases + 1 issueStore read shim removable only after one release cycle ships the (already-landed) eager migration; earliest 1.0.0	LOW
backlog.md	Gitea deferred follow-ups: gh_* agent-tool parity + tea/API create-PR path (tool_github.rs/tool_pull_request.rs GitHub-scoped); richer Gitea Actions/check-runs (degraded to empty); Gitea inline PR-review-comment authoring (viewing works); AI compare-diff for Gitea multi-commit PRs	LOW
backlog.md	Memory P3 (deferred): semantic retrieval only if measured keyword misses justify it (bundled local embeddings + brute-force cosine, no vector DB)	LOW
backlog.md	Rust audit P3: core/mcp_bridge.rs resolve_mcp_name re-spawns every enabled MCP server per mcp_* call — cache advertised-name map at session start	LOW

Doc	Item (still-valid, re-verified 07-30)	Priority
backlog.md	Midway review P3: sweep remaining historical Gemini references (stray descriptive mentions, e.g. src/agents/packetcode.ts description)	LOW
ROADMAP.md	Later: Codex app-server transport (A6); Send-to-Monitor expansion (Approval/Cost, multi-window, PTY attachment); native iOS after PWA; plugin system after distribution unblocks	LOW
ROADMAP.md	Architectural debt: store consolidation review (historyStore/projectHistoryStore/promptStore overlap); mid-session MCP hot-swap unsupported (by design); Remote Agents scope guardrails	LOW
ROADMAP.md	Release-path step 9: expand E2E coverage across workspace session creation, API-agent launch, Remote Agents approval flow, Flight attempt lifecycle	LOW

125 still-valid rows shown · 4 superseded / resolved-in-code / unverifiable items omitted from this rendition (full detail in the HTML report).

10.3 dev/bridgemind — feature-loop ledgers and convergence records

Doc	Item (still-valid, re-verified 07-30)	Priority
packetagent-handoff-loop.md	PH7 gap: PacketAgent W9 publishes no approval-response route, so PacketADE cannot respond to Worker approvals (inspect/pause/resume/revoke landed)	HIGH
packetagent-handoff-loop.md	PH5/PH10: live close-PacketADE/PacketAgent-continues/reconnect continuation gate against a configured live PacketAgent (deploy replay, policy rejection, revoke, network loss, schema skew)	HIGH
dictation-repair-hardening-loop.md	Live microphone acceptance matrix: connect/enable one default + one USB/Bluetooth microphone, run the Settings doctor for each, then exercise the full record/PTT/history/delivery matrix	HIGH
pre-remote-agents-loop-queue.md	Remote Agents remains an unstarted product-decision boundary — the queue explicitly stops there	HIGH
reviewer-gate-loop.md	RG8: packaged local interactive Flight matrix + disposable SSH smoke (disabled/pass/fail/reviewer-error/override/reload paths) to close the loop	MEDIUM
cooperative-flight-graph-loop.md	CG9: release-like packaged local + SSH matrix (landing, reload, cancellation, backward-compat) to close the cooperative-graph loop	MEDIUM
autonomy-policy-loop.md	AP9: adversarial/regression matrix on packaged build + live SSH (cost/time/retry races, policy downgrade, stop behavior) to close the YOLO loop	MEDIUM
coordination-inbox-loop.md	CI9: end-to-end packaged/SSH smoke (user→agent, role broadcast, PTY/MCP, reload, partial failure, YOLO-stop paths)	MEDIUM
flight-supervision-proof-2026-07-28.md	Pickup contract: one isolated packaged local fixture + one disposable SSH fixture must exercise the named RG8/CG9/CI9/AP9 paths before those four items move gated→closed	MEDIUM
flight-supervision-proof-2026-07-28.md	Enabler gap: PacketADE has no production data-directory override, which is the stated blocker for an isolated packaged acceptance run against non-real user Flights	MEDIUM
packetagent-handoff-loop.md	PH3: package builder covers Flight source only — task/attempt and conversation sources plus richer artifact/target inputs pending	MEDIUM
packetagent-handoff-loop.md	PH6: only persisted-cursor polling/ack landed — SSE subscription and richer progress/checkpoint projections pending	MEDIUM
packetagent-handoff-loop.md	PH8: typed artifact return and explicit landing of returned code pending (latest-evidence inspection landed)	MEDIUM
packetagent-handoff-loop.md	PH9: PacketAgent attention events → PacketADE attention queue + coordination inbox integration — still queued, not started	MEDIUM
packetcode-bridgecode-loop.md	PC5: published checksum-verifiable Windows/macOS/Linux PacketCode release channel with clean-machine install/upgrade/rollback proof	MEDIUM
packetcode-bridgecode-loop.md	PC10: cross-repo end-to-end smoke (PacketADE install/detect/configure/launch of a published PacketCode build) + PacketAgent durable-continuation compatibility fixture	MEDIUM
project-local-memory-hub-loop.md	MH8: real external-editor watch storms, partial writes, rename/reload, packaged restart recovery proof	MEDIUM
project-local-memory-hub-loop.md	MH9: opt-in copy/export migration proof plus empty/large/dirty/gitignored packaged-project smoke on available platforms (incl. macOS/Linux hosts, which do not exist yet)	MEDIUM
local-first-mcp-hub-loop.md	MCPH3: unified lifecycle/diagnostics proof against real local/SSH server crash, reload, and version-skew (currently only deterministic tests)	MEDIUM

Doc	Item (still-valid, re-verified 07-30)	Priority
local-first-mcp-hub-loop.md	MCPH8: regression/packaging gate — real Codex CLI + remote-profile trust-proxy parity, offline install/removal, packaged interaction smoke, backward compatibility	MEDIUM
dictation-repair-hardening-loop.md	Packaged cross-platform matrix (DV15 gates): Windows paste/microphone, packaged macOS microphone/accessibility permission, packaged Linux ALSA/PipeWire + X11/Wayland	MEDIUM
trust-provenance-loop.md	TP8: packaged Windows visual/manual inspection, live local/SSH all-provider transport parity (needs live provider credentials + configured SSH host), MCP remote, restart, YOLO, old-state migration matrix; macOS/Linux proof needs those hosts	MEDIUM
pre-remote-agents-loop-queue.md	Signing/updater lane: Windows/macOS signing credentials, notarization, updater configuration for public trusted distribution	MEDIUM
pre-remote-convergence-2026-07-28.md	Live Anthropic multi-turn smoke never went green — two 60s attempts produced no terminal event; the deterministic suite now requires opt-in PACKETADE_LIVE_ANTHROPIC_SMOKE=1 and the live round-trip remains unproven	MEDIUM
pre-remote-convergence-2026-07-28.md	Packaged GitHub + Gitea issue-mirror matrix proof (create/adopt/update/pull/conflict)	MEDIUM
pre-remote-convergence-2026-07-28.md	Packaged multi-display read-only Monitor proof incl. WebView-to-Rust denial integration proof on each platform	MEDIUM
pre-remote-convergence-2026-07-28.md	Distribution gates: signing credentials, notarization, updater config, signed updater manifest (unsigned local artifacts only, with recorded SHA-256s)	MEDIUM
flight-supervision-proof-2026-07-28.md	No disposable SSH acceptance target has been designated (saved credentials/hosts are explicitly not treated as test authorization)	LOW
flight-escalation-loop.md	Open decision 'stuck threshold configurability': fixed 15-min default was chosen; a user-configurable setting (e.g. in OrchestratorSettings) was deferred and never revisited	LOW
packetagent-handoff-loop.md	PH2: richer PacketAgent server/schema capability display in Settings pending (keyring/TLS/health landed)	LOW
packetcode-bridgecode-loop.md	PC9: remaining lower-priority PacketCode hardening loops from the PC8 feature-truth audit	LOW
local-first-mcp-hub-loop.md	Known limitation to eventually close: doctor probe is stdio-only — HTTP/SSE server config is preserved losslessly but gets no live diagnostics	LOW
local-first-mcp-hub-loop.md	Disclosure caveat: a trusted stdio MCP child may still use network per its own runtime — not an OS sandbox (documented posture, no further work scoped)	LOW
dictation-repair-hardening-loop.md	DV17: Parakeet/acceleration/engine benchmarking — intentionally deferred until a packaged CPU Whisper latency/quality baseline is measured with an active microphone; cloud transcription stays off	LOW
pre-remote-convergence-2026-07-28.md	Cosmetic: cargo check carries an existing ts-rs missionId alias warning (intentional Mission→Flight compat alias)	LOW
bridgespace-competitive-brief.md	Strategic takeaway never converted to a task: lead marketing/positioning with the provider/auth-type abstraction + dual transport (the field-unique differentiator) toward the professional-engineer audience	LOW
bridgespace-competitive-brief.md	Paused decision: production-signal monitoring (Sentry/PostHog/CI-to-Flight) is 'not now'; PacketAgent is the eventual owner if reconsidered	LOW
bridgeswarm-teardown.md	Marketing suggestion: keep and market per-attempt worktree isolation as a harder guarantee than BridgeSwarm's advisory file ownership	LOW
packetade-mcp-server-plan.md	Phase 3 ownership-aware MCP tools (claim task ownership, reserve/release file paths) — deferred and never explicitly decided for or against	LOW

39 still-valid rows shown · 11 superseded / resolved-in-code / unverifiable items omitted from this rendition (full detail in the HTML report).

10.4 Subdirectories & archive – remoteagents, moat, mobile, tile-program, zen-workspace, dev/archive

Doc	Item (still-valid, re-verified 07-30)	Priority
09-open-decisions.md	Sprint-0 (a): Auth provider build-vs-buy for v1 (passkey/magic-link on Workers vs Clerk/Auth0/Stytch/Supabase vs Cloudflare Access for internal beta) — owner: Security/Auth agent	CRITICAL

Doc	Item (still-valid, re-verified 07-30)	Priority
09-open-decisions.md	Sprint-0 (b): Payload-encryption launch gate (plaintext internal-only vs required-before-external-private-beta) — owner: project owner + Security/Auth agent	CRITICAL
09-open-decisions.md	Sprint-0 (c): Code location (in-repo remoteagents/ vs apps/ vs separate repo) — owner: implementation lead	CRITICAL
sprints-2026-06-15.md	Unchecked tail item: hold the Remote Agents Sprint-0 kickoff (lock three decisions, scaffold behind compile-off flag)	CRITICAL
README.md	Entire Remote Agents R0 implementation (relay worker, PWA, desktop remote_agents module, six-agent runbook) — zero code exists; the whole package is unexecuted	HIGH
06-implementation-plan.md	All sprints 0-6 unstarted (foundations, relay/presence, snapshots, backend conversation service, mobile UX, push/audit/hardening, beta release)	HIGH
09-open-decisions.md	Backend conversation persistence shape for remote creation (minimal Rust DTO/snapshot for MVP vs fully backend-native post-MVP)	MEDIUM
README.md	Open question 7: does the desktop's pending approval oneshot survive a delayed (minutes-later) phone approval delivered after push? Doc asks for a code check in commands/api_agent.rs before committing	MEDIUM
plan.md	Program status ambiguity + doc hygiene: the tile-program directory carries no COMPLETE/SUPERSEDED banner, is absent from dev/README.md's ownership table, and was not archived in the 2026-07-30 consolidation, while its authoritative spec (conversation-tile-design.md) is archived with a SUPERSEDED ...	MEDIUM
p1-p2-fix-loop-spec.md	Hard kills / force-quit skip RunEvent::Exit leaving orphaned sidecar children; named fast-follow: persistent ~/.packetade/sidecar-active-pids startup-reap registry mirroring reap_orphaned_pty_children — plus detached:true bash shells (openai-agents.ts:932) surviving group-kill and the unix shell...	MEDIUM
09-open-decisions.md	Native iOS strategy (Swift vs React Native vs Capacitor vs Tauri mobile) — explicitly deferred until after PWA beta	LOW
09-open-decisions.md	Deferred-decisions bundle: team/org access model, cloud runner mode, billing model, WebRTC/LAN direct mode, remote PTY policy, mobile file-browser depth, long-term transcript retention	LOW
cost-dashboard-plan.md	Pricing-table maintenance: unknown model IDs price to \$0 (undercounting risk as new models ship)	LOW
cost-dashboard-plan.md	Dashboard UX: date-range filtering and actual-vs-estimated flight cost (plan items 5-6)	LOW
cost-dashboard-plan.md	Per-user cost attribution on shared machines (gap #5)	LOW
plan.md	Post-program: PaneHeaderShell shared-header-grammar extraction	LOW
gitea-support-loop.md	Deferred: Gitea agent-tool (gh_*) parity + tea/API create-PR path; Gitea Actions/check-runs parity (degraded to combined status); AI compare-diff for Gitea multi-commit PRs; GraphQL-only GitHub features stay capability-gated	LOW
memory-v9-loop.md	Deferred: local-embedding semantic retrieval, only if keyword misses are measured	LOW

18 still-valid rows shown · 21 superseded / resolved-in-code / unverifiable items omitted from this rendition (full detail in the HTML report).

Reading the ledger: the dominant pattern is not forgotten work but environment-gated proof — packaged/SSH/live-credential matrices (RG8, CG9, AP9, CI9, TP8, MH8/9, MCPH3/8, S11, dictation microphone, PacketCode/PacketAgent cross-repo gates) that cannot run on this dev host. The genuinely dropped items are rare and now recorded: the sidecar-orphan startup-reap fast-follow from the archived P1/P2 loop, the approval-oneshot-survival check carried over from the mobile research, and the tile-program directory's missing supersession banner.

11. Recommended Next 90 Days

The data supports a clear order: **answer the five owner decisions**, lock in what was fixed, close the trust seams, unblock distribution, then start the frontier bets whose value compounds with time. Do not invert this — every frontier item assumes a foundation that does not silently lose work, every week without CI risks re-breaking the 15 defects just repaired, and every week the decisions sit, 43 verified UX findings and four blocked implementation waves sit with them.

Phase 1 — Lock in & close the trust seams (days 0-14)

- **Answer the five owner decisions + execute Decision 6 (Section 3.1).** Minutes of owner time: D1-D4 have recommended answers; only D5 (Editor vs PacketCode positioning) needs deliberation. Then remove/disable the placebo settings already approved for removal (UX-05/06/07/19). This unblocks all four MS implementation waves and dissolves most of the 15 critical ledger items (Chapter 10).
- **The UX P0 quartet + the two cheap criticals.** UX-01 unscoped inspector, UX-03 preview scoping, UX-04 SSH gating (MS1 correctness boundaries) — alongside the Ctrl+K guard (UX-08) and close-confirmation + view restore (UX-09), two small diffs each.
- **Green the gates and commit.** Run the six verification gates, land the diff, tag it. Nothing proceeds on an unverified base.
- **CI first — the cheapest insurance in this report.** One push-triggered workflow: pnpm lint, vitest run, cargo test. Add clippy + rustfmt while the ladder is open.
- **Close the four open trust items:** Codex "auto" -> workspace-write (bypass only explicit + warning chip); toolUseld on pending_edit; restart_sidecar command; stash-before-remove on failed worktrees.
- **Kill the top three rage-inducers:** terminal-safe shortcut guards, close-confirmation with live counts, selected-view restore.
- **Truth pass:** fix or hide the Prompt Library action and routing card; update CLAUDE.md (v11, GitHub auth, worktree paths, dead conventions).

Phase 2 — Reliability & distribution (days 15-45)

- **Flight recovery:** attempt-aware startup recovery with resume, backend-owned SSH cleanup, worktree/branch GC with auto-exclude — the single biggest reliability gap.
- **Updater, then signing.** The runbook made every decision (~30 lines of code); follow with an OV cert or Azure Trusted Signing; add the compile-only ubuntu/macOS CI matrix in the same window.
- **Streaming performance pass:** memoized rows/renderer, derived chrome selectors, max-interval save flush.
- **Failure visibility:** route logSwallowed and persistence failures through the ToastProvider; one confirm primitive; opt-in crash reports.
- **Flights polish:** sticky launch defaults, per-attempt cost, attempt diff review, notification unification with click-to-jump.
- **Security batch:** backend-constructed ssh argv, prune npx/node grants and the stale CSP entry, SSH trust-state badges, transcript retention + 0600.

Phase 3 — Frontier bets & the remote flagship (days 46-90)

- **Start the two record-now-exploit-later substrates immediately:** the Earned Trust Ledger capture layer + stats panel (#1) and the Flight Recorder journal (#5) — their value compounds with every week of data.
- **Ship Runtime Tripwires' honeypot half (#12)** — the smallest Top 12 item, an active intrusion alarm before autonomy expansion.
- **Remote Agents Sprint 0 folded with the Approval Inbox (#3):** resolve the two blocking decisions (auth provider, encryption timing); protocol package + relay and PWA skeletons + feature flag; inbox desktop-first.

The market brief is unambiguous — and "your own SSH boxes as the fleet" is the differentiated, Anunta-compatible angle.

- **Tournament Mode (#2)** once ledger capture exists — the highest-visible-payoff frontier item.
- **Standards spikes, timeboxed:** MCP 2026-07-28 (Tasks mapped onto Flights), an ACP transport spike on the api-agent:* contract, an OS-sandbox spike (srt/Landlock; Windows options); watch Codex app-server; adopt the Responses API.
- **Defer deliberately:** Rollback Contracts until auto-integration ships; the Adversarial Lane until risk routing exists; SQLite until per-slice files prove insufficient; full mobile until the inbox proves the loop.

The through-line: PacketADE's differentiators — provider breadth, SSH fleets, cost truth, and soon the trust economy no competitor is building — all depend on the same foundation: sessions that survive their own lifecycle, releases that update themselves, and gates that run without willpower. Ninety days of this sequence turns a formidable solo tool into something that can be handed to a second person without apology — precisely the bar the Anunta opportunity and any public beta will demand.